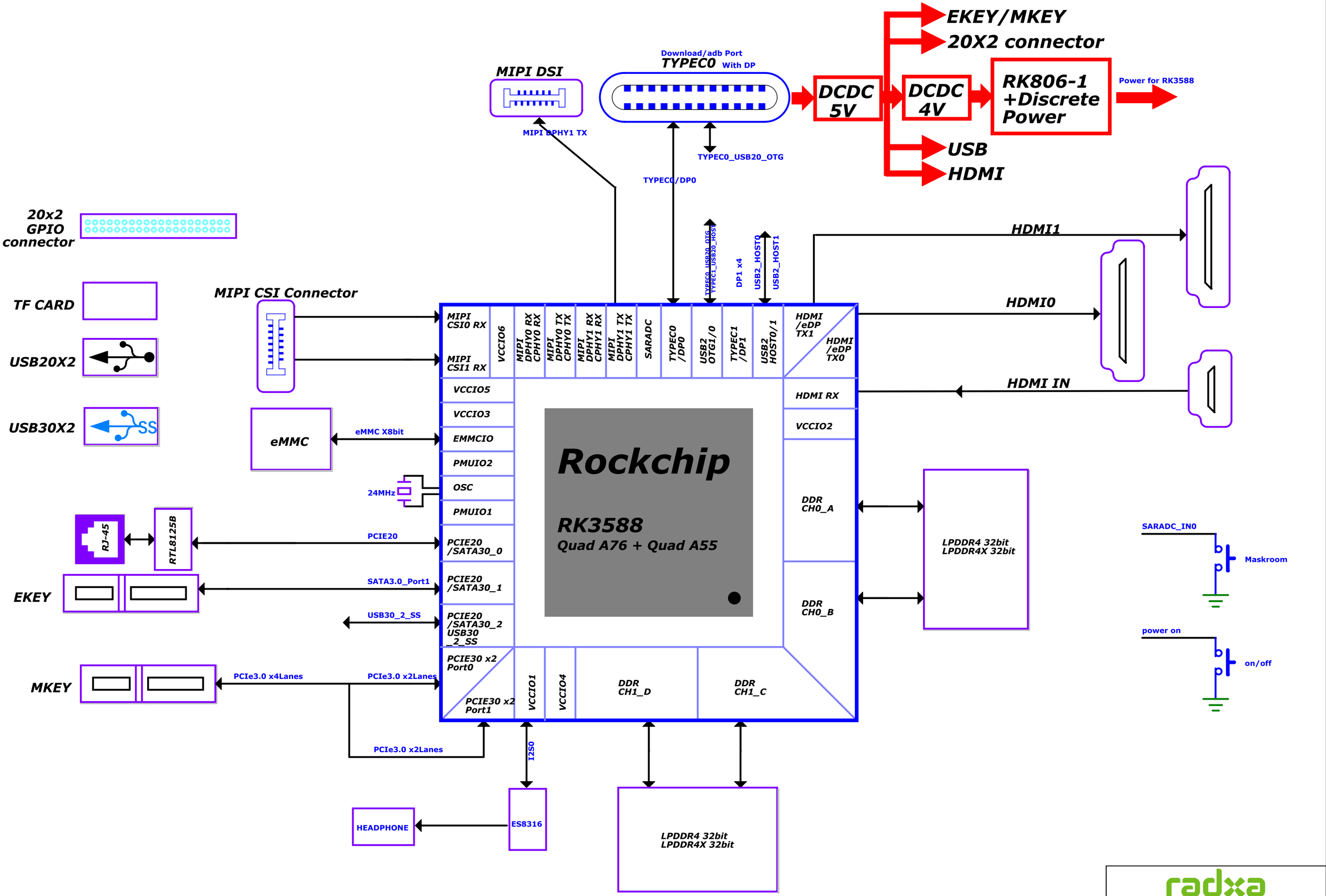
Radxa ROCK 5B+ schematic

V1.2

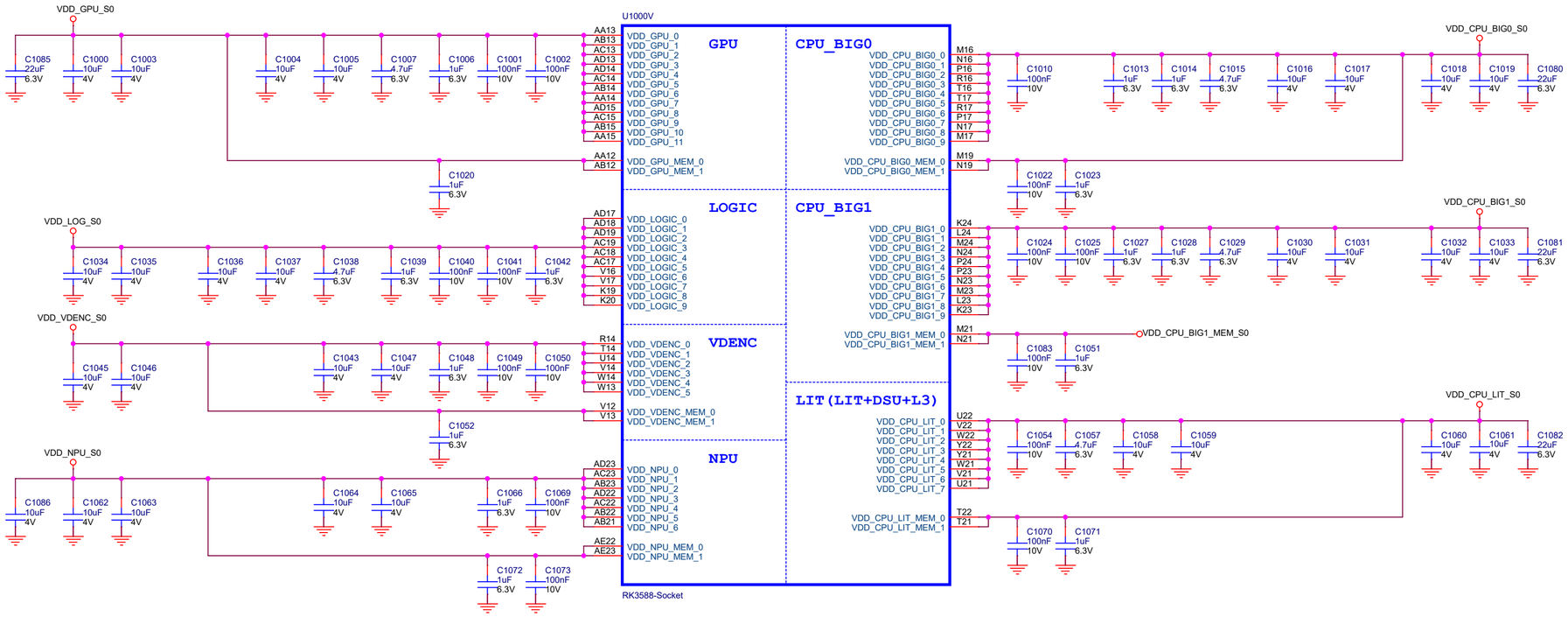
Table of Content

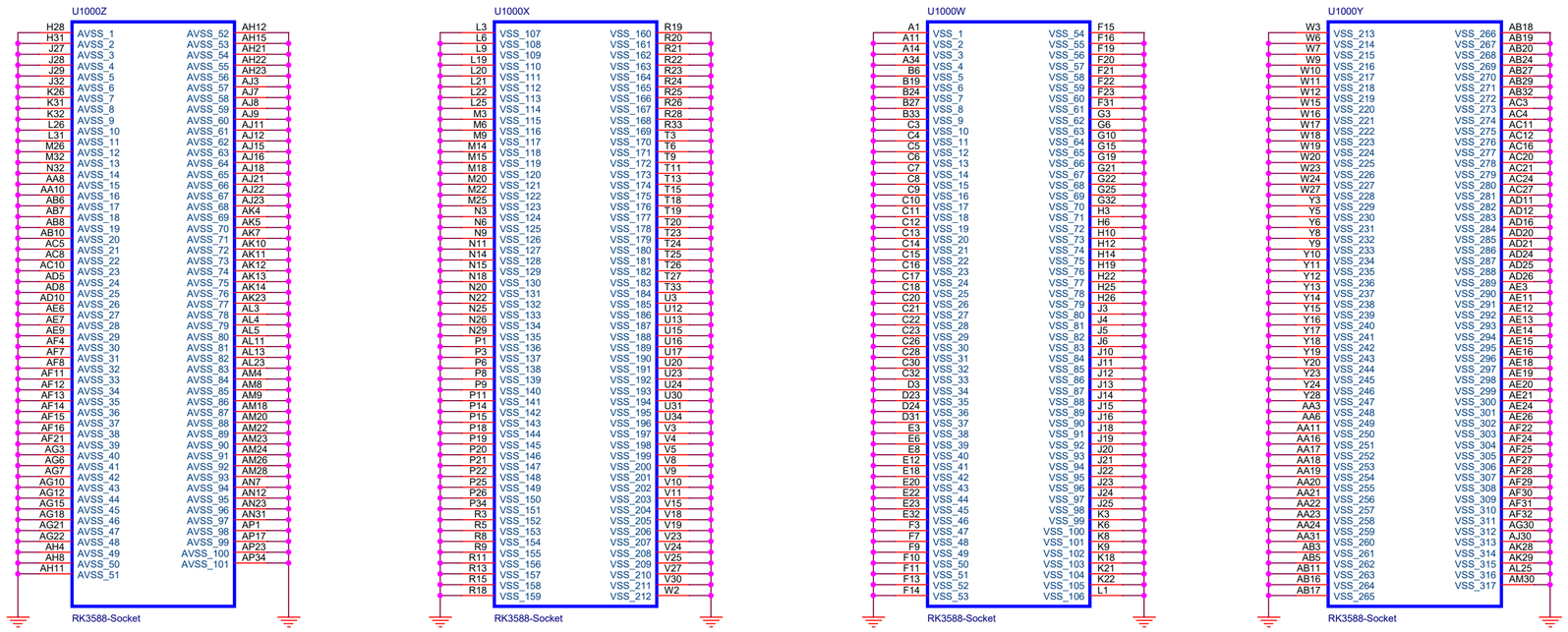
01.Cover
02.Table of Content
03.Block Diagram
04.RK3588_Power/GND
05.RK3588_DDR Controler
06.RK3588_OSC/PLL/PMUIO
07.RK3588_USB30/USB20_Ctrl
08.RK3588_HDMI/eDP Interface
09.RK3588_Flash/SD Controller
10.RK3588_MIPI Interface
11.RK3588_PCIE30/PCIE20/SATA30
12.RK3588_SARADC_GPIO
13.RK3588_ GPIO
14.LPDDR5_315P_2X32bit
15.Flash-TF Card
16.VI-HDMI2.0 RX
17.VO-HDMI2.1 TX
18.Camera_MIPI-CSI
19.VO-LCM_MIPI
20.Audio Codec
21.SPI/eMMC Flash
22.PCIE_WIFI
23.M.2 B key_4G
24.PCIE3.0 M key X2
25.PCIE TO RJ45
26.USB30/20 Port
27.USB20 HUB
28.Type-C Port
29.CONNECT 40pin
30.Power_Ext Discrete
31.Power-PMIC_RK806-1
32.Power_DC 5V
33.Power_TYPEC IN

Size A4	Title: ROCK 5B+	REV 1.2
	Page Name: 02.Table of Content	
Date: Thursday, August 15, 2024	Sheet 2 of 33	

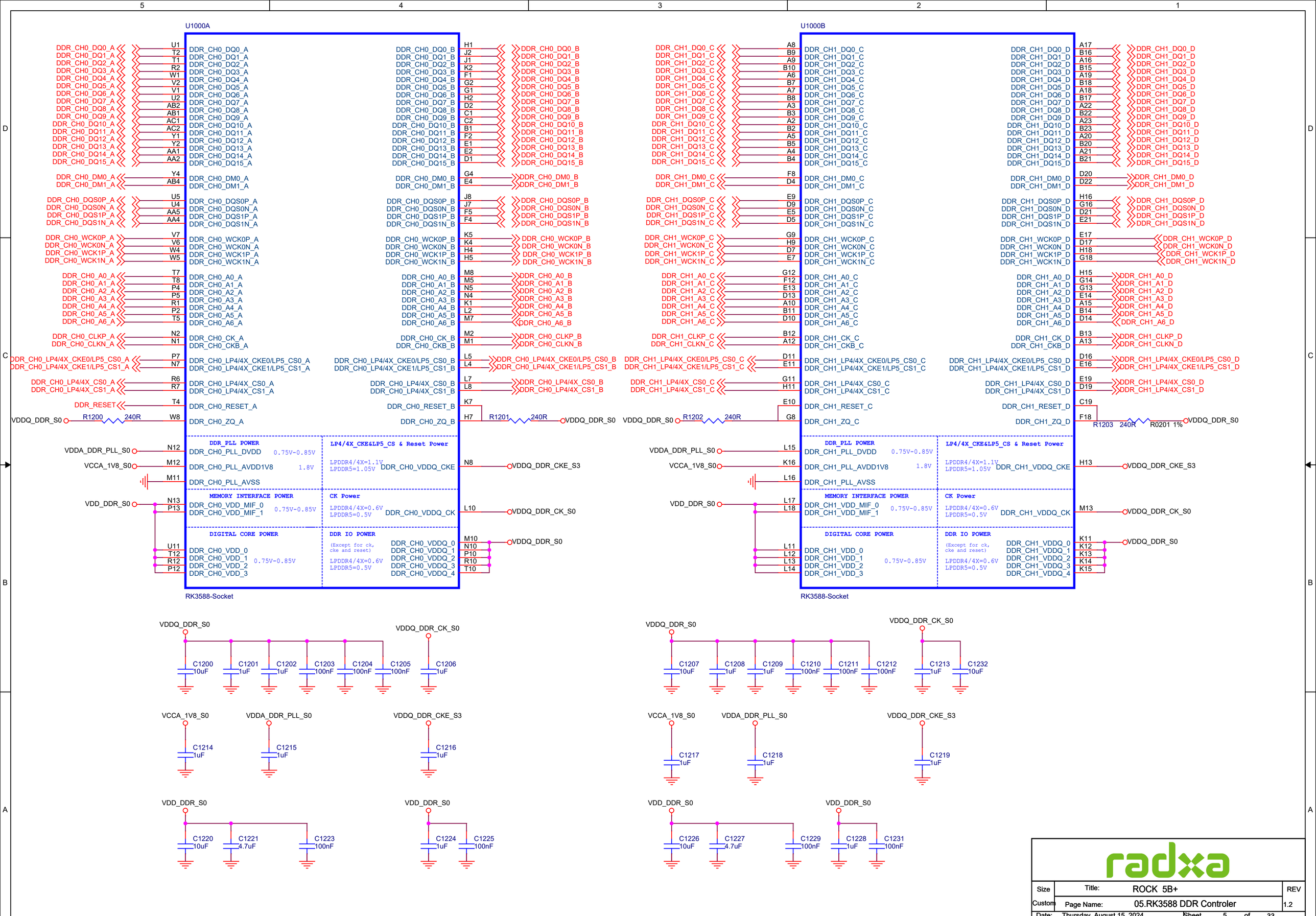


Size A2	Title: ROCK 5B+		REV 1.2
	Page Name: 03.Block Diagram		
Date: Thursday, August 15, 2024		Sheet 3 of 33	





radxa			
Size	Title: ROCK 5B+		REV
A2	Page Name: 04.RK3588_Power/GND		1.2
Date: Thursday, August 15, 2024	Sheet 4	of 33	


U1000A
DDR_CH0_DQ0_A
DDR_CH0_DQ1_A
DDR_CH0_DQ2_A
DDR_CH0_DQ3_A
DDR_CH0_DQ4_A
DDR_CH0_DQ5_A
DDR_CH0_DQ6_A
DDR_CH0_DQ7_A
DDR_CH0_DQ8_A
DDR_CH0_DQ9_A
DDR_CH0_DQ10_A
DDR_CH0_DQ11_A
DDR_CH0_DQ12_A
DDR_CH0_DQ13_A
DDR_CH0_DQ14_A
DDR_CH0_DQ15_A
DDR_CH0_DM0_A
DDR_CH0_DM1_A
DDR_CH0_DQS0P_A
DDR_CH0_DQS0N_A
DDR_CH0_DQS1P_A
DDR_CH0_DQS1N_A
DDR_CH0_WCK0P_A
DDR_CH0_WCK0N_A
DDR_CH0_WCK1P_A
DDR_CH0_WCK1N_A
DDR_CH0_A0_A
DDR_CH0_A1_A
DDR_CH0_A2_A
DDR_CH0_A3_A
DDR_CH0_A4_A
DDR_CH0_A5_A
DDR_CH0_A6_A
DDR_CH0_CLKP_A
DDR_CH0_CLKN_A
DDR_CH0_LP4/4X_CKE0/LP5_CS0_A
DDR_CH0_LP4/4X_CKE1/LP5_CS1_A
DDR_CH0_LP4/4X_CS0_A
DDR_CH0_LP4/4X_CS1_A
DDR_RESET
DDR_CH0_RESET_A
VDDQ_DDR_S0
R1200
240R
DDR_CH0_ZQ_A
DDR_CH0_DQ0_B
DDR_CH0_DQ1_B
DDR_CH0_DQ2_B
DDR_CH0_DQ3_B
DDR_CH0_DQ4_B
DDR_CH0_DQ5_B
DDR_CH0_DQ6_B
DDR_CH0_DQ7_B
DDR_CH0_DQ8_B
DDR_CH0_DQ9_B
DDR_CH0_DQ10_B
DDR_CH0_DQ11_B
DDR_CH0_DQ12_B
DDR_CH0_DQ13_B
DDR_CH0_DQ14_B
DDR_CH0_DQ15_B
DDR_CH0_DM0_B
DDR_CH0_DM1_B
DDR_CH0_DQS0P_B
DDR_CH0_DQS0N_B
DDR_CH0_DQS1P_B
DDR_CH0_DQS1N_B
DDR_CH0_WCK0P_B
DDR_CH0_WCK0N_B
DDR_CH0_WCK1P_B
DDR_CH0_WCK1N_B
DDR_CH0_A0_B
DDR_CH0_A1_B
DDR_CH0_A2_B
DDR_CH0_A3_B
DDR_CH0_A4_B
DDR_CH0_A5_B
DDR_CH0_A6_B
DDR_CH0_CK_B
DDR_CH0_CKB_B
DDR_CH0_LP4/4X_CKE0/LP5_CS0_B
DDR_CH0_LP4/4X_CKE1/LP5_CS1_B
DDR_CH0_LP4/4X_CS0_B
DDR_CH0_LP4/4X_CS1_B
DDR_CH0_RESET_B
DDR_CH0_ZQ_B
R1201
240R
VDDQ_DDR_S0
DDR_PLL POWER
DDR_CH0_PLL_DVDD 0.75V-0.85V
DDR_CH0_PLL_AVDD1V8 1.8V
DDR_CH0_PLL_AVSS
VDDA_DDR_PLL_S0
VCCA_1V8_S0
MEMORY INTERFACE POWER
DDR_CH0_VDD_MIF_0
DDR_CH0_VDD_MIF_1 0.75V-0.85V
VDD_DDR_S0
DIGITAL CORE POWER
DDR_CH0_VDD_0
DDR_CH0_VDD_1 0.75V-0.85V
DDR_CH0_VDD_2
DDR_CH0_VDD_3
LP4/4X_CKE&LP5_CS & Reset Power
LPDDR4/4X=1.1V
LPDDR5=1.05V
DDR_CH0_VDDQ_CKE
VDDQ_DDR_CKE_S3
CK Power
LPDDR4/4X=0.6V
LPDDR5=0.5V
DDR_CH0_VDDQ_CK
VDDQ_DDR_CK_S0
DDR IO POWER
(Except for ck, cke and reset)
LPDDR4/4X=0.6V
LPDDR5=0.5V
DDR_CH0_VDDQ_0
DDR_CH0_VDDQ_1
DDR_CH0_VDDQ_2
DDR_CH0_VDDQ_3
DDR_CH0_VDDQ_4
VDDQ_DDR_S0
RK3588-Socket
U1000B
DDR_CH1_DQ0_C
DDR_CH1_DQ1_C
DDR_CH1_DQ2_C
DDR_CH1_DQ3_C
DDR_CH1_DQ4_C
DDR_CH1_DQ5_C
DDR_CH1_DQ6_C
DDR_CH1_DQ7_C
DDR_CH1_DQ8_C
DDR_CH1_DQ9_C
DDR_CH1_DQ10_C
DDR_CH1_DQ11_C
DDR_CH1_DQ12_C
DDR_CH1_DQ13_C
DDR_CH1_DQ14_C
DDR_CH1_DQ15_C
DDR_CH1_DM0_C
DDR_CH1_DM1_C
DDR_CH1_DQS0P_C
DDR_CH1_DQS0N_C
DDR_CH1_DQS1P_C
DDR_CH1_DQS1N_C
DDR_CH1_WCK0P_C
DDR_CH1_WCK0N_C
DDR_CH1_WCK1P_C
DDR_CH1_WCK1N_C
DDR_CH1_A0_C
DDR_CH1_A1_C
DDR_CH1_A2_C
DDR_CH1_A3_C
DDR_CH1_A4_C
DDR_CH1_A5_C
DDR_CH1_A6_C
DDR_CH1_CLKP_C
DDR_CH1_CLKN_C
DDR_CH1_LP4/4X_CKE0/LP5_CS0_C
DDR_CH1_LP4/4X_CKE1/LP5_CS1_C
DDR_CH1_LP4/4X_CS0_C
DDR_CH1_LP4/4X_CS1_C
DDR_CH1_RESET_C
DDR_CH1_ZQ_C
VDDQ_DDR_S0
R1202
240R
DDR_CH1_DQ0_D
DDR_CH1_DQ1_D
DDR_CH1_DQ2_D
DDR_CH1_DQ3_D
DDR_CH1_DQ4_D
DDR_CH1_DQ5_D
DDR_CH1_DQ6_D
DDR_CH1_DQ7_D
DDR_CH1_DQ8_D
DDR_CH1_DQ9_D
DDR_CH1_DQ10_D
DDR_CH1_DQ11_D
DDR_CH1_DQ12_D
DDR_CH1_DQ13_D
DDR_CH1_DQ14_D
DDR_CH1_DQ15_D
DDR_CH1_DM0_D
DDR_CH1_DM1_D
DDR_CH1_DQS0P_D
DDR_CH1_DQS0N_D
DDR_CH1_DQS1P_D
DDR_CH1_DQS1N_D
DDR_CH1_WCK0P_D
DDR_CH1_WCK0N_D
DDR_CH1_WCK1P_D
DDR_CH1_WCK1N_D
DDR_CH1_A0_D
DDR_CH1_A1_D
DDR_CH1_A2_D
DDR_CH1_A3_D
DDR_CH1_A4_D
DDR_CH1_A5_D
DDR_CH1_A6_D
DDR_CH1_CK_D
DDR_CH1_CKB_D
DDR_CH1_LP4/4X_CKE0/LP5_CS0_D
DDR_CH1_LP4/4X_CKE1/LP5_CS1_D
DDR_CH1_LP4/4X_CS0_D
DDR_CH1_LP4/4X_CS1_D
DDR_CH1_RESET_D
DDR_CH1_ZQ_D
R1203
240R
R0201 1%
VDDQ_DDR_S0
DDR_PLL POWER
DDR_CH1_PLL_DVDD 0.75V-0.85V
DDR_CH1_PLL_AVDD1V8 1.8V
DDR_CH1_PLL_AVSS
VDDA_DDR_PLL_S0
VCCA_1V8_S0
MEMORY INTERFACE POWER
DDR_CH1_VDD_MIF_0
DDR_CH1_VDD_MIF_1 0.75V-0.85V
VDD_DDR_S0
DIGITAL CORE POWER
DDR_CH1_VDD_0
DDR_CH1_VDD_1 0.75V-0.85V
DDR_CH1_VDD_2
DDR_CH1_VDD_3
LP4/4X_CKE&LP5_CS & Reset Power
LPDDR4/4X=1.1V
LPDDR5=1.05V
DDR_CH1_VDDQ_CKE
VDDQ_DDR_CKE_S3
CK Power
LPDDR4/4X=0.6V
LPDDR5=0.5V
DDR_CH1_VDDQ_CK
VDDQ_DDR_CK_S0
DDR IO POWER
(Except for ck, cke and reset)
LPDDR4/4X=0.6V
LPDDR5=0.5V
DDR_CH1_VDDQ_0
DDR_CH1_VDDQ_1
DDR_CH1_VDDQ_2
DDR_CH1_VDDQ_3
DDR_CH1_VDDQ_4
VDDQ_DDR_S0
RK3588-Socket
VDDQ_DDR_S0
VDDQ_DDR_CK_S0
C1200 10uF
C1201 1uF
C1202 1uF
C1203 100nF
C1204 100nF
C1205 100nF
C1206 1uF
VCCA_1V8_S0
VDDA_DDR_PLL_S0
VDDQ_DDR_CKE_S3
C1214 1uF
C1215 1uF
C1216 1uF
VDD_DDR_S0
VDD_DDR_S0
C1220 10uF
C1221 4.7uF
C1223 100nF
C1224 1uF
C1225 100nF
VDDQ_DDR_S0
VDDQ_DDR_CK_S0
C1207 10uF
C1208 1uF
C1209 1uF
C1210 100nF
C1211 100nF
C1212 100nF
C1213 1uF
C1232 10uF
VCCA_1V8_S0
VDDA_DDR_PLL_S0
VDDQ_DDR_CKE_S3
C1217 1uF
C1218 1uF
C1219 1uF
VDD_DDR_S0
VDD_DDR_S0
C1226 10uF
C1227 4.7uF
C1229 100nF
C1228 1uF
C1231 100nF
radxa
Size
Custom
Title: ROCK 5B+
REV
Page Name: 05.RK3588 DDR Controler
1.2
Date: Thursday, August 15, 2024
Sheet 5 of 33

U1000E

OSC

R1101 22R — XOUT24M — T34 XOUT_24M

Y1100 24MHz (1 X1, 2 GND, 3 X2, 4 GND)

R1103 510K

C1101 18pF 50V

XIN24M — R34 XIN_24M

C1102 18pF 50V

VCCA_1V8_S0 — C1103 1uF 4V — C1104 100nF 6.3V — N27 OSC_1V8

PLL

VDD_0V75_S0 — C1105 1uF 4V — C1106 100nF 6.3V — V20 PLL_DVDD0V75

VCCA_1V8_S0 — C1109 100nF 6.3V — U18 PLL_AVDD1V8

U19 PLL_AVSS

PMUIO1 Domain

Operating Voltage=1.8V Only

NPOR_u	M31	RESET_L		
TVSS	M29			
	P33		REFCLK_OUT	GPIO0_A0_d
	P32	TSADC_SHUT_H	TSADC_SHUT_ORG / TSADC_SHUT	GPIO0_A1_z
	R32		PMIC_SLEEP1	GPIO0_A2_d
	R31		PMIC_SLEEP2	GPIO0_A3_d
	P31		SDMMC_DET	GPIO0_A4_u
	N31	SPI2_CLK_M2	PMU_DEBUG / SDMMC_PWREN	GPIO0_A5_d
	N30		I2C0_SDA_M0 / SPI2_MOSI_M2	GPIO0_A6_z
	M30		PMIC_INT_L	GPIO0_A7_u
	L30		UART0_RX_M1 / I2C1_SCL_M1 / SPI2_CS1_M2	GPIO0_B0_z
	K30		PWM5_M0 / UART0_TX_M1 / I2C1_SDA_M1 / SPI2_CS0_M2	GPIO0_B1_z
	K29		CLK32K_OUT0 / CLK32K_IN	GPIO0_B2_u
	L29		I2C0_SCL_M0 / SPI2_MISO_M2	GPIO0_B3_z

RESET_L — R1100 0R — TSADC_SHUT_H

C1100 100nF 6.3V

GPIO0-A0

R1133 22R — PMIC_PWR_CTRL1

CPU_BIG0_VSEL

SDMMC_DET_L

R1104 22R — PMIC_SPI_CLK

R1107 22R — PMIC_SPI_MOSI

PMIC_INT_L

RTC_INT_L

PMIC_SPI_CS

W_DISABLE2

PMUIO1_1V8 — N28 — C1107 100nF 6.3V — VCC1V8_PMU_DDR_S3

PMUIO1/2 Domain Logic Power

Operating Voltage=0.75V

PMU_0V75 — P27 — C1110 100nF 6.3V — C1111 1uF 4V — VDD_0V75_S3

VCC_3V3_S3 — R1139 2.2K — R1140 2.2K — I2C0_SCL_M2 — I2C0_SDA_M2

VCC_3V3_S3 — R95061 2.2K — R95062 2.2K — I2C1_SCL_M2 — I2C1_SDA_M2

RK3588-Socket

U1000F

PMUIO2 Domain

Operating Voltage=1.8V/3.3V

	UART2_TX_M0			I2S1_MCLK_M1	PCIE30X1_1_CLKREQN_M0		I2C1_SCL_M0	JTAG_TCK_M2	GPIO0_B5_d
	UART2_RX_M0			I2S1_SCLK_M1	PCIE30X1_1_WAKEN_M0		I2C1_SDA_M0	JTAG_TMS_M2	GPIO0_B6_d
		CAN0_TX_M0		I2S1_LRCK_M1	PCIE30X1_1_PERSTN_M0	SPI0_CS1_M0	I2C2_SCL_M0	PWM0_M0	GPIO0_B7_d
		CAN0_RX_M0	PDM0_CLK0_M1		PCIE30X1_0_CLKREQN_M0	SPI0_MOSI_M0	I2C2_SDA_M0	PWM1_M0	GPIO0_C0_d
								PMIC_SLEEP3	GPIO0_C1_d
								PMIC_SLEEP4	GPIO0_C2_d
								PMIC_SLEEP5	GPIO0_C3_d
	UART0_RX_M0	DP0_HPDIN_M1	PDM0_CLK1_M1		PCIE30X1_0_WAKEN_M0		I2C4_SDA_M2	PWM2_M0	GPIO0_C4_d
PWM4_M0	UART0_TX_M0	DP1_HPDIN_M1		I2S1_SDI0_M1	PCIE30X1_0_PERSTN_M0		I2C4_SCL_M2	GPU_AVS	GPIO0_C5_u
PWM5_M1	UART0_RTSN	SATA_CP_POD		I2S1_SDI1_M1	PCIE30X4_CLKREQN_M0	SPI0_CLK_M0		NPU_AVS	GPIO0_C6_u
	UART1_RTSN_M2		PDM0_SDI0_M1	I2S1_SDI2_M1	PCIE30X4_WAKEN_M0	SPI0_MISO_M0	I2C6_SDA_M0	PWM6_M0	GPIO0_C7_d
	UART1_CTSN_M2		PDM0_SDI1_M1	I2S1_SDI3_M1	PCIE30X4_PERSTN_M0	SPI3_MISO_M2	I2C6_SCL_M0	PWM7_IR_M0	GPIO0_D0_d
HDMI_TX0_CEC_M1	UART1_TX_M2	UART0_CTSN	HDMI_RX_SDA_M0	I2S1_SDO0_M1	PCIE30X2_CLKREQN_M0	SPI0_CS0_M0	I2C0_SCL_M2	CPU_BIG0_AVS	GPIO0_D1_u
HDMI_TX1_CEC_M1	UART1_RX_M2		HDMI_RX_SCL_M0	I2S1_SDO1_M1	PCIE30X2_WAKEN_M0	SPI3_MOSI_M2	I2C0_SDA_M2		GPIO0_D2_u
						SPI3_CLK_M2		LITCPU_AVS	GPIO0_D3_u
HDMI_TX0_SDA_M1	SATA_CPDET	CAN2_RX_M1	PDM0_SDI2_M1	I2S1_SDO2_M1	PCIE30X2_PERSTN_M0	SPI3_CS0_M2	I2C1_SCL_M2	PWM3_IR_M0	GPIO0_D4_u
HDMI_TX0_SCL_M1	SATA_MP_SWITCH	CAN2_TX_M1		I2S1_SDO3_M1		SPI3_CS1_M2	I2C1_SDA_M2	CPU_BIG1_AVS	GPIO0_D5_u
			PDM0_SDI3_M1					PMIC_SLEEP6	GPIO0_D6_d

P29 UART2_TX_M0

R29 UART2_RX_M0

T28 GPIO0-B7

T31 GPIO0-C0

U32 NPU_VSEL

T32 VSEL4 — R1109 100K — PMIC_PWR_CTRL3

T30 VSEL5 — R1115 100K — PMIC_PWR_CTRL2

R1116 120K — R1110 120K

R30 TYPEC0_SBU1_DC

P30 TYPEC0_SBU2_DC

T29 TP_RST_L

V31 I2C6_SDA_M0

W31 I2C6_SCL_M0

W30 I2C0_SCL_M2

W29 I2C0_SDA_M2

U33 TP_INT_L

V29 I2C1_SCL_M2

V28 I2C1_SDA_M2

W28 CPU_BIG1_VSEL

VCC_3V3_S3 — R1130 2.2K — R1131 2.2K — I2C6_SCL_M0 — I2C6_SDA_M0

PMUIO2_1V8 — R27 — C1112 100nF 6.3V — VCC1V8_PMU_DDR_S3

PMUIO2 — P28 — C1113 100nF 6.3V — VCC_3V3_S3

RK3588-Socket

radxa

Size: Custom	Title: ROCK 5B+		REV: 1.2
	Page Name: 06.RK3588_OSC/PLL/PMUIO		
Date: Thursday, August 15, 2024	Sheet 6 of 33		

U1000M

USB3.0 OTG/DP1.4 Alt of TYPEC0

USB:U3/Gen1----Controller0
DP:RBR/HBR/HBR2/HBR3

TYPEC0_SBU1/DP0_AUXP — AL15 — TYPEC0_SBU1
TYPEC0_SBU2/DP0_AUXN — AM15 — TYPEC0_SBU2

TYPEC0_SSRX1P/DP0_TX0P — AN13 — TYPEC0_SSRX1P
TYPEC0_SSRX1N/DP0_TX0N — AP13 — TYPEC0_SSRX1N

TYPEC0_SSTX1P/DP0_TX1P — AP14 — TYPEC0_SSTX1P
TYPEC0_SSTX1N/DP0_TX1N — AN14 — TYPEC0_SSTX1N

TYPEC0_SSRX2P/DP0_TX2P — AN15 — TYPEC0_SSRX2P
TYPEC0_SSRX2N/DP0_TX2N — AP15 — TYPEC0_SSRX2N

TYPEC0_SSTX2P/DP0_TX3P — AP16 — TYPEC0_SSTX2P
TYPEC0_SSTX2N/DP0_TX3N — AN16 — TYPEC0_SSTX2N

TYPEC0_DP0_REXT — AH16 — TYPEC0_REXT — R1402 8.2K

TYPEC0_DP0_VDD_0V85 — AJ14
TYPEC0_DP0_VDDA_0V85 — AH14 — VDDA_0V85_S0
C1400 1uF 6.3V
C1401 100nF 6.3V

TYPEC0_DP0_VDDH_1V8 — AG14 — VCCA_1V8_S0
C1402 1uF 6.3V
C1403 100nF 6.3V

USB3.0 OTG/DP1.4 Alt of TYPEC1

USB:U3/Gen1----Controller1
DP:RBR/HBR/HBR2/HBR3

TYPEC1_SBU1/DP1_AUXP — AL10
TYPEC1_SBU2/DP1_AUXN — AM10

TYPEC1_SSRX1P/DP1_TX0P — AN8 — TYPEC1_SSRX1P
TYPEC1_SSRX1N/DP1_TX0N — AP8 — TYPEC1_SSRX1N

TYPEC1_SSTX1P/DP1_TX1P — AP9 — TYPEC1_SSTX1P
TYPEC1_SSTX1N/DP1_TX1N — AN9 — TYPEC1_SSTX1N

TYPEC1_SSRX2P/DP1_TX2P — AN10
TYPEC1_SSRX2N/DP1_TX2N — AP10

TYPEC1_SSTX2P/DP1_TX3P — AP11
TYPEC1_SSTX2N/DP1_TX3N — AN11

TYPEC1_DP1_REXT — AG16 — TYPEC1_REXT — R1410 8.2K

TYPEC1_DP1_VDD_0V85 — AH13
TYPEC1_DP1_VDDA_0V85 — AJ13 — VDDA_0V85_S0
C1404 1uF 6.3V
C1405 100nF 6.3V

TYPEC1_DP1_VDDH_1V8 — AG13 — VCCA_1V8_S0
C1406 1uF 6.3V
C1407 100nF 6.3V

RK3588-Socket

U1000L

USB2.0 of TYPEC0 (OTG/HOST/DEVICE)

HS/FS/LS

Download Port

TYPEC0_USB20_OTG_DP — AL12 — TYPEC0_OTG_DP
TYPEC0_USB20_OTG_DM — AM12 — TYPEC0_OTG_DM

TYPEC0_USB20_OTG_ID — AL14
40K
TYPEC0_USB20_VBUSDET — AM14 — TYPEC0_USB20_VBUSDET

TYPEC0_USB20_OTG0_REXT — AP12 — OTG0_REXT — R1405 200R

USB2.0 of TYPEC1 (OTG/HOST/DEVICE)

HS/FS/LS

TYPEC1_USB20_OTG_DP — AK9 — TYPEC1_OTG_DP
TYPEC1_USB20_OTG_DM — AL9 — TYPEC1_OTG_DM

TYPEC1_USB20_OTG_ID — AK8
40K
TYPEC1_USB20_VBUSDET — AL8 — TYPEC0_OTG_VBUSDET

TYPEC1_USB20_OTG1_REXT — AP7 — OTG1_REXT — R1406 200R

USB2.0 HOST0

HS/FS/LS

USB20_HOST0_DP — AK6 — USB20_HOST0_DP
USB20_HOST0_DM — AL6 — USB20_HOST0_DM

USB20_HOST0_REXT — AG9 — HOST0_REXT — R1400 200R

USB2.0 HOST1

HS/FS/LS

USB20_HOST1_DP — AL7 — USB20_HOST1_DP
USB20_HOST1_DM — AM7 — USB20_HOST1_DM

USB20_HOST1_REXT — AH9 — HOST1_REXT — R1401 200R

USB2.0 POWER

USB20_DVDD_0V75 — AH10 — VDDA_0V75_S0
C1408 100nF 6.3V

USB20_AVDD_1V8 — AG11 — VCCA_1V8_S0
C1409 100nF 6.3V

USB20_AVDD_3V3 — AJ10 — VCCA_3V3_S0
C1410 100nF 6.3V

RK3588-Socket

radxa

Size: Custom	Title: ROCK 5B+	REV: 1.2
	Page Name: 07.RK3588_USB30/USB20_Ctrl	
Date: Thursday, August 15, 2024	Sheet 7 of 33	

U1000S

HDMI TX/eDP MUX Port0

HDMI:V2.1 12Gbps
eDP: V1.3 5.4Gbps

HDMI_TX0_SBDP/EDP_TX0_AUXP	AG2	HDMI0_TX_SBDP/eDP0_TX_AUXP	HDMI0_TX_SBDP/eDP0_TX_AUXP
HDMI_TX0_SBDN/EDP_TX0_AUXN	AG1	HDMI0_TX_SBDN/eDP0_TX_AUXN	HDMI0_TX_SBDN/eDP0_TX_AUXN
HDMI_TX0_D0P/EDP_TX0_D0P	AJ2	HDMI0_TX0P_PORT/eDP0_TX_D0P	HDMI0_TX0P_PORT/eDP0_TX_D0P
HDMI_TX0_D0N/EDP_TX0_D0N	AJ1	HDMI0_TX0N_PORT/eDP0_TX_D0N	HDMI0_TX0N_PORT/eDP0_TX_D0N
HDMI_TX0_D1P/EDP_TX0_D1P	AK3	HDMI0_TX1P_PORT/eDP0_TX_D1P	HDMI0_TX1P_PORT/eDP0_TX_D1P
HDMI_TX0_D1N/EDP_TX0_D1N	AK2	HDMI0_TX1N_PORT/eDP0_TX_D1N	HDMI0_TX1N_PORT/eDP0_TX_D1N
HDMI_TX0_D2P/EDP_TX0_D2P	AL2	HDMI0_TX2P_PORT/eDP0_TX_D2P	HDMI0_TX2P_PORT/eDP0_TX_D2P
HDMI_TX0_D2N/EDP_TX0_D2N	AL1	HDMI0_TX2N_PORT/eDP0_TX_D2N	HDMI0_TX2N_PORT/eDP0_TX_D2N
HDMI_TX0_D3P/EDP_TX0_D3P	AH3	HDMI0_TX3P_PORT/eDP0_TX_D3P	HDMI0_TX3P_PORT/eDP0_TX_D3P
HDMI_TX0_D3N/EDP_TX0_D3N	AH2	HDMI0_TX3N_PORT/eDP0_TX_D3N	HDMI0_TX3N_PORT/eDP0_TX_D3N

HDMI/EDP_TX0_REXT — AM2 HDMI_TX0_REXT — R1708 8.2K 1%

HDMI/EDP_TX0_VDD_0V75 — AA9 — VDDA_0V75_S0
- C1715 4.7uF 10V
- C1751 4.7uF 10V

HDMI/EDP_TX0_AVDD_0V75 — AB9
- C1714 1uF 6.3V

HDMI/EDP_TX0_VDD_IO_1V8 / HDMI/EDP_TX0_VDD_CMN_1V8 — AC7, AC6 — VCC_1V8_S0
- C1716 1uF 6.3V
- C1717 100nF 6.3V
- C1742 4.7uF 10V

HDMI TX/eDP MUX Port1

HDMI:V2.1 12Gbps
eDP: V1.3 5.4Gbps

HDMI_TX1_SBDP/EDP_TX1_AUXP	AN2	HDMI1_TX_SBDP/eDP1_TX_AUXP	HDMI1_TX_SBDP/eDP1_TX_AUXP
HDMI_TX1_SBDN/EDP_TX1_AUXN	AP2	HDMI1_TX_SBDN/eDP1_TX_AUXN	HDMI1_TX_SBDN/eDP1_TX_AUXN
HDMI_TX1_D0P/EDP_TX1_D0P	AN4	HDMI1_TX0P_PORT/eDP1_TX_D0P	HDMI1_TX0P_PORT/eDP1_TX_D0P
HDMI_TX1_D0N/EDP_TX1_D0N	AP4	HDMI1_TX0N_PORT/eDP1_TX_D0N	HDMI1_TX0N_PORT/eDP1_TX_D0N
HDMI_TX1_D1P/EDP_TX1_D1P	AM5	HDMI1_TX1P_PORT/eDP1_TX_D1P	HDMI1_TX1P_PORT/eDP1_TX_D1P
HDMI_TX1_D1N/EDP_TX1_D1N	AN5	HDMI1_TX1N_PORT/eDP1_TX_D1N	HDMI1_TX1N_PORT/eDP1_TX_D1N
HDMI_TX1_D2P/EDP_TX1_D2P	AN6	HDMI1_TX2P_PORT/eDP1_TX_D2P	HDMI1_TX2P_PORT/eDP1_TX_D2P
HDMI_TX1_D2N/EDP_TX1_D2N	AP6	HDMI1_TX2N_PORT/eDP1_TX_D2N	HDMI1_TX2N_PORT/eDP1_TX_D2N
HDMI_TX1_D3P/EDP_TX1_D3P	AM3	HDMI1_TX3P_PORT/eDP1_TX_D3P	HDMI1_TX3P_PORT/eDP1_TX_D3P
HDMI_TX1_D3N/EDP_TX1_D3N	AN3	HDMI1_TX3N_PORT/eDP1_TX_D3N	HDMI1_TX3N_PORT/eDP1_TX_D3N

HDMI/EDP_TX1_REXT — AN1 HDMI_TX1_REXT — R1719 8.2K

HDMI/EDP_TX1_VDD_0V75 — AD9 — VDDA_0V75_S0
- C1735 1uF 6.3V
- C1752 100nF 10V
- C1730 4.7uF 10V

HDMI/EDP_TX1_AVDD_0V75 — AC9
- C1729 1uF 6.3V

HDMI/EDP_TX1_VDD_IO_1V8 / HDMI/EDP_TX1_VDD_CMN_1V8 — AD7, AD6 — VCC_1V8_S0
- C1731 1uF 6.3V
- C1732 100nF 6.3V
- C1733 4.7uF 10V

RK3588-Socket

U1000T

HDMI RX

HDMI:V2.0

HDMI_RX_D0P	AG5	HDMI_RX_D0P_IN
HDMI_RX_D0N	AG4	HDMI_RX_D0N_IN
HDMI_RX_D1P	AH6	HDMI_RX_D1P_IN
HDMI_RX_D1N	AH5	HDMI_RX_D1N_IN
HDMI_RX_D2P	AJ5	HDMI_RX_D2P_IN
HDMI_RX_D2N	AJ4	HDMI_RX_D2N_IN
HDMI_RX_CLKP	AF6	HDMI_RX_CLKP_IN
HDMI_RX_CLKN	AF5	HDMI_RX_CLKN_IN

HDMI_RX_REXT — AF3 — R1730 200R

HDMI_RX_AVDD0V75 — AE8 — VDDA_0V75_S0
- C1738 1uF 6.3V

HDMI_RX_VPH3V3 — AE4
HDMI_RX_DVDD3V3 — AE5 — VCCA_3V3_S0
- C1740 100nF 6.3V
- C1741 1uF 6.3V

RK3588-Socket

Size	Title:	ROCK 5B+	REV
Custom	Page Name:	08.RK3588_HDMI/eDP Interface	1.2
	Date:	Thursday, August 15, 2024	Sheet 8 of 33

U1000C

EMMCIO Domain

Operating Voltage=1.8V

FSPI_D0_M0 / EMMC_D0	/ GPIO2_D0_u	Y33
FSPI_D1_M0 / EMMC_D1	/ GPIO2_D1_u	W33
FSPI_D2_M0 / EMMC_D2	/ GPIO2_D2_u	V32
FSPI_D3_M0 / EMMC_D3	/ GPIO2_D3_u	AA33
UART5_RX_M2 / I2C1_SCL_M3 / EMMC_D4	/ GPIO2_D4_u	Y32
UART5_TX_M2 / I2C1_SDA_M3 / EMMC_D5	/ GPIO2_D5_u	AA32
FSPI_CS0N_M0/ EMMC_D6	/ GPIO2_D6_u	W32
FSPI_CS1N_M0/ EMMC_D7	/ GPIO2_D7_u	V33
FSPI_CLK_M0 / EMMC_CMD	/ GPIO2_A0_u	W34
/ EMMC_CLKOUT	/ GPIO2_A1_d	V34
UART5_CTSN_M1 / I2C2_SDA_M2 / EMMC_DATA_STROBE	/ GPIO2_A2_d	Y34
UART5_RTSN_M1 / I2C2_SCL_M2 / EMMC_RSTN	/ GPIO2_A3_d	AA34
EMMCIO_1V8		V26

RK3588-Socket

Y33 — eMMC_D0
W33 — eMMC_D1
V32 — eMMC_D2
AA33 — eMMC_D3
Y32 — eMMC_D4
AA32 — eMMC_D5
W32 — eMMC_D6
V33 — eMMC_D7

W34 — eMMC_CMD
V34 — R1317 0R — eMMC_CLKOUT
Y34 — eMMC_DATA_STROBE
AA34 — eMMC_RSTn

C145 NC 16V

V26 — VCC_1V8_S3

C1300 1uF 4V

U1000D

VCCIO2 Domain

Operating Voltage=1.8V/3.3V

PWM8_M1 / I2C3_SCL_M4 / PDM1_SDI3_M0 / JTAG_TCK_M1	/ UART2_TX_M1 / SDMMC_D0	/ GPIO4_D0_u
PWM9_M1 / I2C3_SDA_M4 / PDM1_SDI2_M0 / JTAG_TMS_M1	/ UART2_RX_M1 / SDMMC_D1	/ GPIO4_D1_u
/ I2C8_SCL_M0 / PDM1_SDI1_M0 / JTAG_TCK_M0	/ UART5_CTSN_M0 / SDMMC_D2	/ GPIO4_D2_u
PWM10_M1 / I2C8_SDA_M0 / PDM1_SDI0_M0 / JTAG_TMS_M0	/ UART5_RTSN_M0 / SDMMC_D3	/ GPIO4_D3_u
PWM7_IR_M1 / CAN0_TX_M1 / PDM1_CLK1_M0 / MCU_JTAG_TCK_M0	/ UART5_RX_M0 / SDMMC_CMD	/ GPIO4_D4_u
TEST_CLKOUT_M0 / CAN0_RX_M1 / PDM1_CLK0_M0 / MCU_JTAG_TMS_M0	/ UART5_TX_M0 / SDMMC_CLK	/ GPIO4_D5_d
VCCIO2_1V8		AA7
VCCIO2		Y7

RK3588-Socket

AD2 — SDMMC0_D0
AD1 — SDMMC0_D1
AF2 SDMMC_D2/JTAG_TCK_M0 — SDMMC_D2/JTAG_TCK_M0
AF1 SDMMC_D3/JTAG_TMS_M0 — SDMMC_D3/JTAG_TMS_M0
AE2 SDMMC_CMD/MCU_JTAG_TCK_M0 — SDMMC_CMD/MCU_JTAG_TCK_M0
AE1 SDMMC_CLK/MCU_JTAG_TMS_M0 — SDMMC_CLK/MCU_JTAG_TMS_M0

AA7 — VCC_1V8_S0

C1301 100nF 6.3V

Y7 — VCCIO_SD_S0

C1302 100nF 6.3V

radxa

Size A4	Title: ROCK 5B+	REV 1.2
	Page Name: 09.RK3588_Flash/SD Controller	
Date: Thursday, August 15, 2024	Sheet 9 of 33	

U1000Q

MIPI D/C-PHY DSI_TX Port0

D-PHY:V2.0 4.5Gbps/Lane
C-PHY:V1.1 5.7Gbps/Trio

MIPI_DPHY0_TX_CLKP/MIPI_CPHY0_TX_TRIO1_C	AN26	
MIPI_DPHY0_TX_CLKN/MIPI_CPHY0_TX_TRIO1_B	AP26	
MIPI_DPHY0_TX_D0P/MIPI_CPHY0_TX_TRIO0_B	AN24	
MIPI_DPHY0_TX_D0N/MIPI_CPHY0_TX_TRIO0_A	AP24	
MIPI_DPHY0_TX_D1P/MIPI_CPHY0_TX_TRIO1_A	AN25	
MIPI_DPHY0_TX_D1N/MIPI_CPHY0_TX_TRIO0_C	AP25	
MIPI_DPHY0_TX_D2P/MIPI_CPHY0_TX_TRIO2_B	AN27	
MIPI_DPHY0_TX_D2N/MIPI_CPHY0_TX_TRIO2_A	AP27	
MIPI_DPHY0_TX_D3P/NO_USE	AN28	
MIPI_DPHY0_TX_D3N/MIPI_CPHY0_TX_TRIO2_C	AP28	

MIPI D/C-PHY CSI_RX Port0

D-PHY:V2.0 4.5Gbps/Lane
C-PHY:V1.1 5.7Gbps/Trio

MIPI_DPHY0_RX_CLKP/MIPI_CPHY0_RX_TRIO1_C	AN32	
MIPI_DPHY0_RX_CLKN/MIPI_CPHY0_RX_TRIO1_B	AP31	
MIPI_DPHY0_RX_D0P/MIPI_CPHY0_RX_TRIO0_B	AN29	
MIPI_DPHY0_RX_D0N/MIPI_CPHY0_RX_TRIO0_A	AP29	
MIPI_DPHY0_RX_D1P/MIPI_CPHY0_RX_TRIO1_A	AN30	
MIPI_DPHY0_RX_D1N/MIPI_CPHY0_RX_TRIO0_C	AP30	
MIPI_DPHY0_RX_D2P/MIPI_CPHY0_RX_TRIO2_B	AN33	
MIPI_DPHY0_RX_D2N/MIPI_CPHY0_RX_TRIO2_A	AP32	
MIPI_DPHY0_RX_D3P/NO_USE	AN34	
MIPI_DPHY0_RX_D3N/MIPI_CPHY0_RX_TRIO2_C	AP33	

MIPI D/C-PHY POWER

MIPI_D/C_PHY0_VREG	AF20	C1600 2.2nF6.3V	
MIPI_D/C_PHY0_VDD	AG20	C1602 100nF 6.3V; C1601 1uF 6.3V	VDD_0V75_S0
MIPI_D/C_PHY0_VDD_1V2	AH20	C1603 1uF 6.3V	VDDA_1V2_S0
MIPI_D/C_PHY0_VDD_1V8	AJ20	C1605 100nF 10V	VCC_1V8_S0

RK3588-Socket
U1000R

MIPI D/C-PHY DSI_TX Port1

D-PHY:V2.0 4.5Gbps/Lane
C-PHY:V1.1 5.7Gbps/Trio

MIPI_DPHY1_TX_CLKP/MIPI_CPHY1_TX_TRIO1_C	AN20	MIPI_DPHY1_TX_CLKP
MIPI_DPHY1_TX_CLKN/MIPI_CPHY1_TX_TRIO1_B	AP20	MIPI_DPHY1_TX_CLKN
MIPI_DPHY1_TX_D0P/MIPI_CPHY1_TX_TRIO0_B	AN18	MIPI_DPHY1_TX_D0P
MIPI_DPHY1_TX_D0N/MIPI_CPHY1_TX_TRIO0_A	AP18	MIPI_DPHY1_TX_D0N
MIPI_DPHY1_TX_D1P/MIPI_CPHY1_TX_TRIO1_A	AN19	MIPI_DPHY1_TX_D1P
MIPI_DPHY1_TX_D1N/MIPI_CPHY1_TX_TRIO0_C	AP19	MIPI_DPHY1_TX_D1N
MIPI_DPHY1_TX_D2P/MIPI_CPHY1_TX_TRIO2_B	AN21	MIPI_DPHY1_TX_D2P
MIPI_DPHY1_TX_D2N/MIPI_CPHY1_TX_TRIO2_A	AP21	MIPI_DPHY1_TX_D2N
MIPI_DPHY1_TX_D3P/NO_USE	AN22	MIPI_DPHY1_TX_D3P
MIPI_DPHY1_TX_D3N/MIPI_CPHY1_TX_TRIO2_C	AP22	MIPI_DPHY1_TX_D3N

MIPI D/C-PHY CSI_RX Port1

D-PHY:V2.0 4.5Gbps/Lane
C-PHY:V1.1 5.7Gbps/Trio

MIPI_DPHY1_RX_CLKP/MIPI_CPHY1_RX_TRIO1_C	AK20	
MIPI_DPHY1_RX_CLKN/MIPI_CPHY1_RX_TRIO1_B	AL20	
MIPI_DPHY1_RX_D0P/MIPI_CPHY1_RX_TRIO0_B	AK18	
MIPI_DPHY1_RX_D0N/MIPI_CPHY1_RX_TRIO0_A	AL18	
MIPI_DPHY1_RX_D1P/MIPI_CPHY1_RX_TRIO1_A	AK19	
MIPI_DPHY1_RX_D1N/MIPI_CPHY1_RX_TRIO0_C	AL19	
MIPI_DPHY1_RX_D2P/MIPI_CPHY1_RX_TRIO2_B	AK21	
MIPI_DPHY1_RX_D2N/MIPI_CPHY1_RX_TRIO2_A	AL21	
MIPI_DPHY1_RX_D3P/NO_USE	AK22	
MIPI_DPHY1_RX_D3N/MIPI_CPHY1_RX_TRIO2_C	AL22	

MIPI D/C-PHY POWER

MIPI_D/C_PHY1_VREG	AF19	C1608 2.2nF 6.3V	
MIPI_D/C_PHY1_VDD	AG19	C1609 1uF 6.3V	VDD_0V75_S0
MIPI_D/C_PHY1_VDD_1V2	AH19	C1611 100nF 6.3V	VDDA_1V2_S0
MIPI_D/C_PHY1_VDD_1V8	AJ19	C1613 1uF 6.3V	VCC_1V8_S0

RK3588-Socket

U1000P

MIPI DPHY CSI_RX Port0

MIPI V1.2/2.5Gbps

MIPI_CSI0_CLK0P	AJ33	MIPI_CSI0_RX_CLK0P
MIPI_CSI0_CLK0N	AJ34	MIPI_CSI0_RX_CLK0N
MIPI_CSI0_D0P	AG33	MIPI_CSI0_RX_D0P
MIPI_CSI0_D0N	AG34	MIPI_CSI0_RX_D0N
MIPI_CSI0_D1P	AH33	MIPI_CSI0_RX_D1P
MIPI_CSI0_D1N	AH34	MIPI_CSI0_RX_D1N
MIPI_CSI0_CLK1P	AM33	MIPI_CSI0_RX_CLK1P
MIPI_CSI0_CLK1N	AM34	MIPI_CSI0_RX_CLK1N
MIPI_CSI0_D2P	AK33	MIPI_CSI0_RX_D2P
MIPI_CSI0_D2N	AK34	MIPI_CSI0_RX_D2N
MIPI_CSI0_D3P	AL33	MIPI_CSI0_RX_D3P
MIPI_CSI0_D3N	AL34	MIPI_CSI0_RX_D3N
MIPI_CSI0_AVCC0V75	AB25	C1616 100nF 6.3V; C1615 1uF 6.3V; VDD_0V75_S0
MIPI_CSI0_AVCC1V8	AB26	C1617 1uF 6.3V; VCC_1V8_S0

MIPI DPHY CSI_RX Port1

MIPI V1.2/2.5Gbps

MIPI_CSI1_CLK0P	AJ31	MIPI_CSI1_RX_CLK0P
MIPI_CSI1_CLK0N	AJ32	MIPI_CSI1_RX_CLK0N
MIPI_CSI1_D0P	AG31	MIPI_CSI1_RX_D0P
MIPI_CSI1_D0N	AG32	MIPI_CSI1_RX_D0N
MIPI_CSI1_D1P	AH31	MIPI_CSI1_RX_D1P
MIPI_CSI1_D1N	AH32	MIPI_CSI1_RX_D1N
MIPI_CSI1_CLK1P	AM31	MIPI_CSI1_RX_CLK1P
MIPI_CSI1_CLK1N	AM32	MIPI_CSI1_RX_CLK1N
MIPI_CSI1_D2P	AK31	MIPI_CSI1_RX_D2P
MIPI_CSI1_D2N	AK32	MIPI_CSI1_RX_D2N
MIPI_CSI1_D3P	AL31	MIPI_CSI1_RX_D3P
MIPI_CSI1_D3N	AL32	MIPI_CSI1_RX_D3N
MIPI_CSI1_AVCC0V75	AA25	C1619 1uF 6.3V; VDD_0V75_S0
MIPI_CSI1_AVCC1V8	AA26	C1621 100nF 6.3V; VCC_1V8_S0

RK3588-Socket

radxa

Size	Title:	ROCK 5B+	REV
Custom	Page Name:	10.RK3588_MIPI Interface	1.2
	Date:	Thursday, August 15, 2024	Sheet 10 of 33

U1000N

PCIE20/SATA30 Mux0

PCIe20x1_2
1Lane(RC)--4(1L2)

SATA30 HOST
Controller0

MUX

input or output

PCIE20_0_REFCLKP PCIE20_0_REFCLKN	L32 L33	PCIE20_0_REFCLKP PCIE20_0_REFCLKN
PCIE20_0_TXP/SATA30_0_TXP PCIE20_0_TXN/SATA30_0_TXN	M34 M33	PCIE20_0_TXP PCIE20_0_TXN
PCIE20_0_RXP/SATA30_0_RXP PCIE20_0_RXN/SATA30_0_RXN	N33 N34	PCIE20_0_RXP PCIE20_0_RXN
PCIE20_SATA30_0_AVDD_0V85	M28	VDDA_0V85_S0
		C1810 10uF 4V
PCIE20_SATA30_0_AVDD_1V8	M27	VCCA_1V8_S0
		C1808 1uF 4V

PCIE20:Gen1/Gen2
SATA30:Gen1/Gen2/Gen3

PCIE20/SATA30 Mux1

PCIe30x1_0
1Lane(RC)--2(1L0)

SATA30 HOST
Controller1

MUX

input or output

PCIE20_1_REFCLKP PCIE20_1_REFCLKN	H32 H33	PCIE20_1_REFCLKP PCIE20_1_REFCLKN
PCIE20_1_TXP/SATA30_1_TXP PCIE20_1_TXN/SATA30_1_TXN	K33 K34	PCIE20_1_TXP PCIE20_1_TXN
PCIE20_1_RXP/SATA30_1_RXP PCIE20_1_RXN/SATA30_1_RXN	J33 J34	PCIE20_1_RXP PCIE20_1_RXN
PCIE20_SATA30_1_AVDD_0V85	L28	VDDA_0V85_S0
		C1818 100nF 6.3V
PCIE20_SATA30_1_AVDD_1V8	L27	VCCA_1V8_S0
		C1816 100nF 6.3V

PCIE20:Gen1/Gen2
SATA30:Gen1/Gen2/Gen3

PCIE20/SATA30/USB30 HOST Mux2

PCIe30x1_1
1Lane(RC)--3(1L1)

SATA30 HOST
Controller2

USB30 HOST
Controller2

MUX

input or output

PCIE20_2_REFCLKP PCIE20_2_REFCLKN	G31 G30	
PCIE20_2_TXP/SATA30_2_TXP/USB30_2_SSTXP PCIE20_2_TXN/SATA30_2_TXN/USB30_2_SSTXN	H30 H29	USB30_2_SSTXP USB30_2_SSTXN
PCIE20_2_RXP/SATA30_2_RXP/USB30_2_SSRXP PCIE20_2_RXN/SATA30_2_RXN/USB30_2_SSRXN	J31 J30	USB30_2_SSRXP USB30_2_SSRXN
PCIE20_SATA30_USB30_2_AVDD_0V85	K28	VDDA_0V85_S0
		C1806 1uF 4V
PCIE20_SATA30_USB30_2_AVDD_1V8	K27	VCCA_1V8_S0
		C1804 1uF 4V

PCIE20:Gen1/Gen2
USB 30:Gen1
SATA30:Gen1/Gen2/Gen3

RK3588-Socket

U1000O

PCIE30X4

PCIe30X4
4Lane(DM)--0(4L)

PCIe30x1_0
1Lane(RC)--2(1L0)

PCIE20/SATA30 MUX1

PCIE30:
PCIe1.1(2.5Gbps)
PCIe2.1(5Gbps)
PCIe3.0(8Gbps)

PCIe30x2
2Lane(RC)--1(2L)

PCIe30x1_1
1Lane(RC)--3(1L1)

PCIE20/SATA30/USB30 MUX2

PCIE30 PORT0

Lane0
Lane1

input

PCIE30_PORT0_REF_CLKP PCIE30_PORT0_REF_CLKN	E33 E34	PCIE30_PORT0_REFCLKP_IN PCIE30_PORT0_REFCLKN_IN
PCIE30_PORT0_TX0P PCIE30_PORT0_TX0N	D32 D33	PCIE30_PORT0_TX0P PCIE30_PORT0_TX0N
PCIE30_PORT0_RX0P PCIE30_PORT0_RX0N	G33 G34	PCIE30_PORT0_RX0P PCIE30_PORT0_RX0N
PCIE30_PORT0_TX1P PCIE30_PORT0_TX1N	C33 C34	PCIE30_PORT0_TX1P PCIE30_PORT0_TX1N
PCIE30_PORT0_RX1P PCIE30_PORT0_RX1N	F32 F33	PCIE30_PORT0_RX1P PCIE30_PORT0_RX1N
PCIE30_PORT0_RESREF	B34	PCIE30_0_REF R1801 200R
PCIE30_PORT0_AVDD0V75	G24	VDD_0V75_S0
		C1802 1uF 6.3V C1803 4.7uF 10V
PCIE30_PORT0_AVDD1V8	G23	VCCA_1V8_S0
		C1800 1uF 6.3V C1801 4.7uF 10V

PCIE30 PORT1

Lane0
Lane1

input

PCIE30_PORT1_REF_CLKP PCIE30_PORT1_REF_CLKN	A28 B28	PCIE30_PORT1_REFCLKP_IN PCIE30_PORT1_REFCLKN_IN
PCIE30_PORT1_TX0P PCIE30_PORT1_TX0N	B30 A30	PCIE30_PORT1_TX2P PCIE30_PORT1_TX2N
PCIE30_PORT1_RX0P PCIE30_PORT1_RX0N	B32 A32	PCIE30_PORT1_RX2P PCIE30_PORT1_RX2N
PCIE30_PORT1_TX1P PCIE30_PORT1_TX1N	C29 B29	PCIE30_PORT1_TX3P PCIE30_PORT1_TX3N
PCIE30_PORT1_RX1P PCIE30_PORT1_RX1N	C31 B31	PCIE30_PORT1_RX3P PCIE30_PORT1_RX3N
PCIE30_PORT1_RESREF	A33	PCIE30_1_REF R1807 200R
PCIE30_PORT1_AVDD0V75	H24	VDD_0V75_S0
		C1814 1uF 6.3V C1815 4.7uF 10V
PCIE30_PORT1_AVDD1V8	H23	VCCA_1V8_S0
		C1812 100nF 6.3V C1813 4.7uF 10V

RK3588-Socket

radxa

Size: Custom	Title: ROCK 5B+		REV: 1.2
	Page Name: 11.RK3588_PCIE30/PCIE20/SATA30		
Date: Thursday, August 15, 2024		Sheet 11 of 33	

RK3588_H(VCCIO3 Domain)

U1000G

VCCIO1 Domain

Operating Voltage=1.8V

SPI4_MISO_M0	UART3_RX_M0		I2C3_SDA_M0			GPIO1_C0_z
SPI4_MOSI_M0	UART3_TX_M0		I2C3_SCL_M0			GPIO1_C1_z
SPI4_CLK_M0	UART3_RTSN	PWM3_IR_M2	I2C6_SDA_M1		I2S0_MCLK	GPIO1_C2_d
SPI4_CS0_M0	UART3_CTSN	PWM7_IR_M2	I2C6_SCL_M1		I2S0_SCLK	GPIO1_C3_d
SPI4_CS1_M0		PWM11_IR_M2	I2C2_SDA_M3	PDM0_CLK1_M0		GPIO1_C4_d
	UART4_RTSN		I2C2_SCL_M3		I2S0_LRCK	GPIO1_C5_d
		PWM15_IR_M2	I2C4_SDA_M4	PDM0_CLK0_M0		GPIO1_C6_d
	UART4_CTSN		I2C4_SCL_M4		I2S0_SDO0	GPIO1_C7_d
SPI1_MISO_M2	UART6_TX_M2		I2C7_SCL_M0		I2S0_SDO1	GPIO1_D0_d
SPI1_MOSI_M2	UART6_RX_M2		I2C7_SDA_M0	PDM0_SDI1_M0	I2S0_SDO2/I2S0_SDI3	GPIO1_D1_d
SPI1_CLK_M2	UART4_TX_M0	PWM0_M1	I2C1_SCL_M4	PDM0_SDI2_M0	I2S0_SDO3/I2S0_SDI2	GPIO1_D2_d
SPI1_CS0_M2	UART4_RX_M0	PWM1_M1	I2C1_SDA_M4	PDM0_SDI3_M0	I2S0_SDI1	GPIO1_D3_d
					I2S0_SDI0	GPIO1_D4_d
SPI1_CS1_M2				PDM0_SDI0_M0		GPIO1_D5_d

G29	GPIO1_C0_z	I2C3_SDA_M0_MIPI
G27	GPIO1_C1_z	I2C3_SCL_M0_MIPI
F30	GPIO1_C2_d	I2S0_MCLK
E31	GPIO1_C3_d	I2S0_SCLK_TX
E30	GPIO1_C4_d	LCD_PWREN_H
D30	GPIO1_C5_d	I2S0_LRCK_TX
D29	GPIO1_C6_d	HDMIIRX_DET_L
E29	GPIO1_C7_d	I2S0_SDO0
F26	GPIO1_D0_d	I2C7_SCL_M0_CODEC
F27	GPIO1_D1_d	I2C7_SDA_M0_CODEC
F28	GPIO1_D2_d	EKEY_EN
E28	GPIO1_D3_d	HP_INT
D28	GPIO1_D4_d	I2S0_SDI0
G26	GPIO1_D5_d	HP_DETECT

VCCIO1_1V8: G20 — C1502 100nF 6.3V — VCC_1V8_S0

VCC_1V8_S0: R1506 2.2K, R1507 2.2K, R1501 2.2K, R1502 2.2K

RK3588-Socket

U1000U

SARADC

12-bit 1MS/s

SARADC_IN0_BOOT	AM16	BOOT_SARADC_IN0
Recovery/SARADC_IN1	AL16	SARADC_VIN1_KEY/RECOVERY
SARADC_IN2	AK16	SARADC_VIN2_BOM_ID
SARADC_IN3	AN17	SARADC_VIN3_HP_HOOK
SARADC_IN4	AM17	SARADC_IN4
SARADC_IN5	AK15	SARADC_VIN5_EVB_HW_ID
SARADC_IN6	AL17	SARADC_VIN6_dcin
SARADC_IN7	AK17	SARADC_VIN7_dcin
SARADC_AVDD_1V8	AH18	VCCA_1V8_S0

C1501 100nF 6.3V

OTP

NC	AD4	
OTP_VDDOTP_0V75	AD3	VDD_0V75_S0

C1505 100nF 6.3V

TSADC

TSADC_TEST_OUT_TS	AF18	

RK3588-Socket

VCCA_1V8_S0: R95027 10K, R1500 10K

C1513 1nF, C1511 1nF, C1510 1nF, C1512 1nF

SW9208 TS018B

ED15 BTR04G02

U1000H

VCCIO3 Domain

Operating Voltage=1.8V

	UART6_RX_M0	FSPI_D0_M1	SDIO_D0_M0	GMAC0_RXD2		GPIO2_A6_u
	UART6_TX_M0	FSPI_D1_M1	SDIO_D1_M0	GMAC0_RXD3		GPIO2_A7_u
I2C8_SCL_M1	UART6_RTSN_M0	FSPI_D2_M1	SDIO_D2_M0	GMAC0_RXCLK		GPIO2_B0_u
I2C8_SDA_M1	UART6_CTSN_M0	FSPI_D3_M1	SDIO_D3_M0	GMAC0_TXD2		GPIO2_B1_u
I2C3_SCL_M3			SDIO_CMD_M0	GMAC0_TXD3		GPIO2_B2_u
I2C3_SDA_M3		FSPI_CLK_M1	SDIO_CLK_M0	GMAC0_TXCLK		GPIO2_B3_d
I2C4_SDA_M1	UART7_RX_M0	FSPI_CS0N_M1	HDMI_TX1_SDA_M0	GMAC0_PTP_REFCLK		GPIO2_B4_u
I2C4_SCL_M1	UART7_TX_M0	FSPI_CS1N_M1	HDMI_TX1_SCL_M0	GMAC0_PPSTRIG		GPIO2_B5_u
I2C5_SCL_M4	UART1_RX_M0		I2S2_MCLK_M0	GMAC0_TXD0		GPIO2_B6_d
I2C5_SDA_M4	UART1_TX_M0		I2S2_SCLK_M0	GMAC0_TXD1		GPIO2_B7_d
I2C2_SDA_M1	UART1_RTSN_M0	SPI1_CLK_M0	I2S2_LRCK_M0	GMAC0_TXEN		GPIO2_C0_d
I2C2_SCL_M1	UART1_CTSN_M0	SPI1_MISO_M0		GMAC0_RXD0		GPIO2_C1_d
I2C6_SDA_M2	UART9_TX_M0	SPI1_MOSI_M0		GMAC0_RXD1		GPIO2_C2_d
I2C6_SCL_M2		SPI1_CS0_M0	I2S2_SDI_M0	ETH0_REFCLKO_25M		GPIO2_C3_d
TEST_CLKOUT_M1	UART9_RX_M0	SPI1_CS1_M0	HDMI_TX1_CEC_M0	GMAC0_PPSCLK		GPIO2_C4_d
				CLK32K_OUT1		GPIO2_C5_d
	UART7_RTSN_M0	SPI3_CS0_M0	PWM2_M2	GMAC0_RXDV_CRS		GPIO4_C2_d
I2C7_SCL_M1	PWM4_M1	SPI3_CS1_M0	I2S2_SDO_M0	GMAC0_MCLKINOUT		GPIO4_C3_d
I2C7_SDA_M1	UART9_RTSN_M0	SPI3_MISO_M0	PWM5_M2	GMAC0_MDC		GPIO4_C4_d
I2C0_SCL_M1	UART9_CTSN_M0	SPI3_MOSI_M0	PWM6_M2	GMAC0_MDIO		GPIO4_C5_d
I2C0_SDA_M1	UART7_CTSN_M0	SPI3_CLK_M0	PWM7_IR_M3	GMAC0_TXER		GPIO4_C6_d

AC32	GPIO2_A6_u	MIPI_CSI1_PDN2_H
AC31	GPIO2_A7_u	MIPI_CSI1_PDN4_H
AE32	GPIO2_B0_u	CAM2_RST_L
AC33	GPIO2_B1_u	4G_PWREN
AC34	GPIO2_B2_u	4G_WAKE_ON_HOST
AE33	GPIO2_B3_d	4G_RESET
AB31	GPIO2_B4_u	I2C4_SDA
AB30	GPIO2_B5_u	I2C4_SCL
AD33	GPIO2_B6_d	TYPEC5V_PWREN_H
AD34	GPIO2_B7_d	I2S2_SCLK_M0_BT
AE34	GPIO2_C0_d	I2S2_LRCK_M0_BT
AD32	GPIO2_C1_d	LCD_RESET
AD31	GPIO2_C2_d	LCD_light_EN
AD30	GPIO2_C3_d	I2S2_SDI_M0_BT
AC30	GPIO2_C4_d	HDMITX1_CEC_M2
AE30	GPIO2_C5_d	XRSTJ
AE31	GPIO4_C2_d	LCD_PWM
AF34	GPIO4_C3_d	I2S2_SDO_M0_BT
AB34	GPIO4_C4_d	GPIO4-C4
AB33	GPIO4_C5_d	GPIO4-C5
AF33	GPIO4_C6_d	GPIO4-C6

R119 NC

VCCIO3_1V8: Y26 — C1503 100nF 6.3V — VCC_1V8_S0

VCCA_1V8_S0: R95029 10K, R95030 10K

I2C4_SDA, I2C4_SCL

RK3588-Socket

VCCA_1V8_S0, VCC_3V3_S0

GND — C90407 100nF

U90081

B2	VCCA	VCCB	A2
A3	A1	B1	A1
B3	A2	B2	B1
C3	A3	B3	C1
D3	A4	B4	D1
D2	GND	OE	C2

UM3204

I2C4_SDA — I2C4_SDA-3V3

I2C4_SCL — I2C4_SCL-3V3

GPIO4-C5 — GPIO4-C5/

GPIO4-C6 — GPIO4-C6/

GND, VCCA_1V8_S0

VCCA_1V8_S0: R1511 100K, R1508 100K, R165 NC

KEY — J9510 (2, 1)

BOOT_SARADC_IN0

SARADC_VIN5_EVB_HW_ID

SARADC_VIN2_BOM_ID

R1509 DNP, R1510 NC, R166 100K

radxa

Size: Custom	Title: ROCK 5B+		REV: 1.2
	Page Name: 12.RK3588_SARADC_GPIO		
Date: Thursday, August 15, 2024	Sheet 12 of 33		

U1000J

VCCIO5 Domain

Operating Voltage=1.8V/3.3V

PWM10_M0	SPI4_MISO_M1		I2C6_SDA_M4	FSPI_D0_M2	I2S3_MCLK	SDIO_D0_M1	GMAC1_TXD2
AUDDSM_LN	SPI4_MOSI_M1	PWM11_IR_M0	I2C6_SCL_M4	FSPI_D1_M2	I2S3_SCLK	SDIO_D1_M1	GMAC1_TXD3
AUDDSM_LP	SPI4_CLK_M1	UART8_TX_M1		FSPI_D2_M2	I2S3_LRCK	SDIO_D2_M1	GMAC1_RXD2
AUDDSM_RN	SPI4_CS0_M1	UART8_RX_M1		FSPI_D3_M2	I2S3_SDO	SDIO_D3_M1	GMAC1_RXD3
AUDDSM_RP	SPI4_CS1_M1	UART8_RTSN_M1			I2S3_SDI	SDIO_CMD_M1	GMAC1_TXCLK
	MIPI_CAMERA0_CLK_M1	UART8_CTSN_M1	I2C4_SDA_M0	FSPI_CLK_M2		SDIO_CLK_M1	GMAC1_RXCLK
	MIPI_CAMERA1_CLK_M1		I2C4_SCL_M0				ETH1_REFCLKO_25M
PWM8_M0	MIPI_CAMERA2_CLK_M1						GMAC1_RXD0
PWM9_M0	MIPI_CAMERA3_CLK_M1						GMAC1_RXD1
PWM2_M1	MIPI_CAMERA4_CLK_M1	UART2_TX_M2					GMAC1_RXDV_CRS
PWM3_IR_M1		UART2_RX_M2			I2S2_SDI_M1		GMAC1_TXER
		UART2_RTSN			I2S2_SDO_M1		GMAC1_TXD0
		UART2_CTSN			I2S2_MCLK_M1		GMAC1_TXD1
PWM12_M0	CAN1_RX_M0	UART3_TX_M1			I2S2_SCLK_M1		GMAC1_TXEN
PWM13_M0	CAN1_TX_M0	UART3_RX_M1			I2S2_LRCK_M1		GMAC1_MCLKINOUT
			I2C3_SCL_M1	SPI1_MOSI_M1	HDMI_TX1_HPD_M1		GMAC1_PTP_REF_CLK
		UART7_TX_M1	I2C3_SDA_M1	SPI1_MISO_M1			GMAC1_PPSTRIG
	PCIE30X2_BUTTON_RSTN	UART7_RX_M1		SPI1_CLK_M1			GMAC1_PPSCLK
PWM14_M0		UART7_RTSN_M1	I2C8_SCL_M4	SPI1_CS0_M1		MIPI_TE0	GMAC1_MDC
PWM15_IR_M0		UART7_CTSN_M1	I2C8_SDA_M4	SPI1_CS1_M1		MIPI_TE1	GMAC1_MDIO
CAN2_RX_M0	PCIE30X4_CLKREQN_M2	UART5_TX_M1	FSPI_CS0N_M2	SPI3_CS0_M3	HDMI_TX1_CEC_M2		CIF_D8
CAN2_TX_M0	PCIE30X4_WAKEN_M2	UART5_RX_M1	FSPI_CS1N_M2	SPI3_CS1_M3	HDMI_TX1_SDA_M1		CIF_D9
	PCIE30X4_PERSTN_M2			SPI3_MISO_M3	HDMI_TX1_SCL_M1		CIF_D10
	PCIE20X1_2_CLKREQN_M0		I2C5_SCL_M0	SPI3_MOSI_M3	HDMI_TX0_SCL_M2		CIF_D11
PWM8_M2	PCIE20X1_2_WAKEN_M0	UART4_RX_M1	I2C5_SDA_M0	SPI3_CLK_M3	HDMI_TX0_SDA_M2		CIF_D12
PWM9_M2	PCIE20X1_2_PERSTN_M0	UART4_TX_M1		SPI0_MISO_M3	HDMI_RX_CEC_M1		CIF_D13
	PCIE30X2_CLKREQN_M2	UART9_RTSN_M2	I2C7_SCL_M2	SPI0_MOSI_M3	HDMI_RX_SCL_M1		CIF_D14
PWM10_M2	PCIE30X2_WAKEN_M2	UART9_CTSN_M2	I2C7_SDA_M2	SPI0_CLK_M3	HDMI_RX_SDA_M1		CIF_D15
	PCIE30X2_PERSTN_M2	UART9_RX_M2		SPI0_CS0_M3	HDMI_RX_HPDOUT_M1	HDMI_TX0_HPD_M1	MCU_JTAG_TCK_M1
PWM11_IR_M3	PCIE30X4_BUTTON_RSTN	UART9_TX_M2		SPI0_CS1_M3	DP1_HPDIN_M0		MCU_JTAG_TMS_M1

GPIO3_A0_u	AA29	FSPI-D0
GPIO3_A1_u	AA30	FSPI-D1
GPIO3_A2_u	AD27	FSPI-D2
GPIO3_A3_u	AE27	FSPI-D3
GPIO3_A4_d	AD28	UART8_RTSn_M1_BT
GPIO3_A5_d	AH30	FSPI-CLK
GPIO3_A6_d	AH27	4G_DISABLE
GPIO3_A7_u	AG29	PWM8_M0
GPIO3_B0_u	AG28	PCIE_PERST_L
GPIO3_B1_d	AH29	I2S2_LRCK_RX_M1
GPIO3_B2_d	AE28	I2S2_SDI_M1
GPIO3_B3_u	AC28	I2S2_SDO_M1
GPIO3_B4_u	AC29	CC_INT_L
GPIO3_B5_u	AD29	I2S2_SCLK_TX_M1
GPIO3_B6_d	AE29	I2S2_LRCK_TX_M1
GPIO3_B7_d	AA28	SPI1_MOSI_M1
GPIO3_C0_d	Y29	SPI1_MISO_M1
GPIO3_C1_d	Y27	SPI1_CLK_M1
GPIO3_C2_d	Y31	SPI1_CS0_M1
GPIO3_C3_d	Y30	SPI1_CS1_M1
GPIO3_C4_u	AH26	FSPI-CS0
GPIO3_C5_u	AH25	HDMITX1_SDA_M1
GPIO3_C6_u	AG26	HDMITX1_SCL_M1
GPIO3_C7_u	AJ24	PCIE20_1_2_CLKREQn_M1_L
GPIO3_D0_u	AH24	PCIE20_1_2_WAKEn_M1_L
GPIO3_D1_d	AG23	HDMI_RX_CEC
GPIO3_D2_d	AG25	HDMI_RX_SCL_M1
GPIO3_D3_d	AG24	HDMI_RX_SDA_M1
GPIO3_D4_d	AA27	HDMIIRX_HPDOUT_H
GPIO3_D5_d	AB28	BT_WAKE_L

VCCIO5_1V8 — W25 — VCC_1V8_S0

C1903 100nF 6.3V

VCCIO5 — W26 — VCC_3V3_S0

C1902 100nF 6.3V

RK3588-Socket

U1000K

VCCIO6 Domain

Operating Voltage=1.8V/3.3V

	SPI0_MISO_M1	UART9_RTSN_M1		I2S1_MCLK_M0	PCIE30X1_1_CLKREQN_M1	BT1120_D0	CIF_D0	GPIO4_A0_d
	SPI0_MOSI_M1	UART9_CTSN_M1		I2S1_SCLK_M0	PCIE30X1_1_WAKEN_M1	BT1120_D1	CIF_D1	GPIO4_A1_d
	SPI0_CLK_M1			I2S1_LRCK_M0	PCIE30X1_1_PERSTN_M1	BT1120_D2	CIF_D2	GPIO4_A2_d
		UART0_TX_M2			PCIE30X1_0_CLKREQN_M1	BT1120_D3	CIF_D3	GPIO4_A3_d
	SPI2_MISO_M1	UART0_RX_M2	I2C3_SCL_M2		PCIE30X1_0_WAKEN_M1	BT1120_D4	CIF_D4	GPIO4_A4_d
	SPI2_MOSI_M1	UART3_TX_M2	I2C3_SDA_M2	I2S1_SDI0_M0	PCIE30X1_0_PERSTN_M1	BT1120_D5	CIF_D5	GPIO4_A5_d
	SPI2_CLK_M1	UART3_RX_M2	I2C5_SCL_M2	I2S1_SDI1_M0	PCIE30X2_CLKREQN_M1	BT1120_D6	CIF_D6	GPIO4_A6_d
	SPI2_CS0_M1		I2C5_SDA_M2	I2S1_SDI2_M0	PCIE30X2_WAKEN_M1	BT1120_D7	CIF_D7	GPIO4_A7_d
	SPI2_CS1_M1	UART8_TX_M0	I2C6_SDA_M3	I2S1_SDI3_M0	PCIE30X2_PERSTN_M1	BT1120_CLKOUT	CIF_CLKIN	GPIO4_B0_d
SATA2_ACT_LED_M0	SPDIF1_TX_M1	SPI0_CS1_M1	UART8_RX_M0	I2C6_SCL_M3	I2S1_SDO0_M0	PCIE30X1_0_BUTTON_RSTN		MIPI_CAMERA0_CLK_M0	GPIO4_B1_u
CAN1_RX_M1	PWM14_M1	SPI0_CS0_M1	UART8_RTSN_M0	I2C7_SCL_M3	I2S1_SDO1_M0	PCIE30X1_1_BUTTON_RSTN	BT1120_D8	CIF_HREF	GPIO4_B2_u
CAN1_TX_M1	PWM15_IR_M1		UART8_CTSN_M0	I2C7_SDA_M3	I2S1_SDO2_M0	PCIE20X1_2_BUTTON_RSTN	BT1120_D9	CIF_VSYNC	GPIO4_B3_u
SPDIF0_TX_M1	PWM11_IR_M1	DP0_HPDIN_M0	UART9_TX_M1		I2S1_SDO3_M0	PCIE30X4_CLKREQN_M1	BT1120_D10	CIF_CLKOUT	GPIO4_B4_u
SATA1_ACT_LED_M0	PWM12_M1	SPI3_MISO_M1	UART9_RX_M1		HDMI_RX_CEC_M0	PCIE30X4_WAKEN_M1	BT1120_D11		GPIO4_B5_d
SATA0_ACT_LED_M0	PWM13_M1	SPI3_MOSI_M1		I2C5_SCL_M1	HDMI_RX_HPDOUT_M0	PCIE30X4_PERSTN_M1	BT1120_D12		GPIO4_B6_d
		SPI3_CLK_M1		I2C5_SDA_M1	HDMI_TX0_SCL_M0	PCIE20X1_2_CLKREQN_M1	BT1120_D13		GPIO4_B7_u
		SPI3_CS0_M1		I2C8_SCL_M3	HDMI_TX0_SDA_M0	PCIE20X1_2_WAKEN_M1	BT1120_D14		GPIO4_C0_u
SPDIF1_TX_M2	PWM6_M1	SPI3_CS1_M1		I2C8_SDA_M3	HDMI_TX0_CEC_M0	PCIE20X1_2_PERSTN_M1	BT1120_D15		GPIO4_C1_d

GPIO4_A0_d	AK30	CM_RST_L
GPIO4_A1_d	AL30	HDMI1_TX_ON_H
GPIO4_A2_d	AM29	W-DISABLE
GPIO4_A3_d	AL29	PCIE30x1_0_CLKREQn_M1_L
GPIO4_A4_d	AL28	PCIE30x1_0_WAKEn_M1_L
GPIO4_A5_d	AK27	PCIE30x1_0_PERSTn_M1_L
GPIO4_A6_d	AL27	PCIE30x2_CLKREQn_M1
GPIO4_A7_d	AM27	PCIE30x2_WAKEn_M1
GPIO4_B0_d	AK26	PCIE30x2_PERSTn_M1
GPIO4_B1_u	AL24	HDMI0_TX_ON_H
GPIO4_B2_u	AK25	I2C7_SCL_M3
GPIO4_B3_u	AM25	I2C7_SDA_M3
GPIO4_B4_u	AL26	PCIE30X4_CLKREQn_M1_L
GPIO4_B5_d	AJ26	PCIE30X4_WAKEn_M1_L
GPIO4_B6_d	AJ27	PCIE30X4_PERSTn_M1_L
GPIO4_B7_u	AJ28	HDMITX0_SCL_M0
GPIO4_C0_u	AJ25	HDMITX0_SDA_M0
GPIO4_C1_d	AK24	HDMITX0_CEC_M0

VCCIO6_1V8 — AC25 — VCC_1V8_S0

C1909 100nF 6.3V

VCCIO6 — AC26 — VCC_3V3_S0

C1908 100nF 6.3V

RK3588-Socket

U1000I

VCCIO4 Domain

Operating Voltage=1.8V/3.3V

		SPI4_MISO_M2	PCIE30X1_1_CLKREQN_M2	DP0_HPDIN_M2		I2C2_SDA_M4	UART6_RX_M1	GPIO1_A0_d
SATA1_ACT_LED_M1		SPI4_MOSI_M2	PCIE30X1_1_WAKEN_M2	DP1_HPDIN_M2		I2C2_SCL_M4	UART6_TX_M1	GPIO1_A1_d
	PWM0_M2	SPI4_CLK_M2		VOP_POST_EMPTY		I2C4_SDA_M3	UART6_RTSN_M1	GPIO1_A2_d
	PWM1_M2	SPI4_CS0_M2		HDMI_TX1_SDA_M2		I2C4_SCL_M3	UART6_CTSN_M1	GPIO1_A3_d
		SPI2_MISO_M0		HDMI_TX1_SCL_M2				GPIO1_A4_d
		SPI2_MOSI_M0		HDMI_TX0_HPD_M0				GPIO1_A5_d
		SPI2_CLK_M0		HDMI_TX1_HPD_M0				GPIO1_A6_d
	PWM3_IR_M3	SPI2_CS0_M0	PCIE30X1_1_PERSTN_M2			PDM1_SDI0_M1		GPIO1_A7_u
		SPI2_CS1_M0	PCIE30X4_CLKREQN_M3			PDM1_SDI1_M1		GPIO1_B0_u
		SPI0_MISO_M2	PCIE30X4_WAKEN_M3			PDM1_SDI2_M1		GPIO1_B1_d
		SPI0_MOSI_M2	PCIE30X4_PERSTN_M3			PDM1_SDI3_M1	UART4_RX_M2	GPIO1_B2_d
SATA0_ACT_LED_M1		SPI0_CLK_M2	PCIE30X1_0_WAKEN_M2			PDM1_CLK1_M1	UART4_TX_M2	GPIO1_B3_d
		SPI0_CS0_M2	PCIE30X1_0_PERSTN_M2			PDM1_CLK0_M1	UART7_RX_M2	GPIO1_B4_u
		SPI0_CS1_M2	PCIE30X1_0_CLKREQN_M2				UART7_TX_M2	GPIO1_B5_u
	HDMI_RX_HPDOUT_M2	SPDIF0_TX_M0	PCIE30X2_WAKEN_M3	MIPI_CAMERA1_CLK_M0		I2C5_SCL_M3	UART1_TX_M1	GPIO1_B6_u
SATA2_ACT_LED_M1	HDMI_RX_CEC_M2	PWM13_M2	SPDIF1_TX_M0	PCIE30X2_PERSTN_M3	MIPI_CAMERA2_CLK_M0	I2C5_SDA_M3	UART1_RX_M1	GPIO1_B7_u
	HDMI_RX_SCL_M2	PWM14_M2			MIPI_CAMERA3_CLK_M0	I2C8_SCL_M2	UART1_RTSN_M1	GPIO1_D6_u
	HDMI_RX_SDA_M2	PWM15_IR_M3		PCIE30X2_CLKREQN_M3	MIPI_CAMERA4_CLK_M0	I2C8_SDA_M2	UART1_CTSN_M1	GPIO1_D7_u

GPIO1_A0_d	A24	DEVSLP
GPIO1_A1_d	A25	USB_HOST_PWREN_H
GPIO1_A2_d	A26	GPIO1_A2
GPIO1_A3_d	A27	GPIO1_A3
GPIO1_A4_d	B25	PCIE30x4_PWREN_H
GPIO1_A5_d	B26	HDMITX0_HPDIN_M1
GPIO1_A6_d	C24	HDMITX1_HPDIN_M1
GPIO1_A7_u	C25	MIPI_CSI0_PDN1_H
GPIO1_B0_u	C27	MIPI_CSI0_PDN0_H
GPIO1_B1_d	D25	SPI0-MISO
GPIO1_B2_d	D26	SPI0-MOSI
GPIO1_B3_d	D27	SPI0-CLK
GPIO1_B4_u	E24	SPI0-CS0
GPIO1_B5_u	E25	SPI0-CS1
GPIO1_B6_u	E26	MIPI_CAM1_CLKOUT
GPIO1_B7_u	E27	MIPI_CAM2_CLKOUT
GPIO1_D6_u	F24	MIPI_CAM3_CLKOUT
GPIO1_D7_u	F25	MIPI_CAM4_CLKOUT

VCCIO4_1V8 — H20 — VCC_1V8_S0

C1907 100nF 6.3V

VCCIO4 — H21

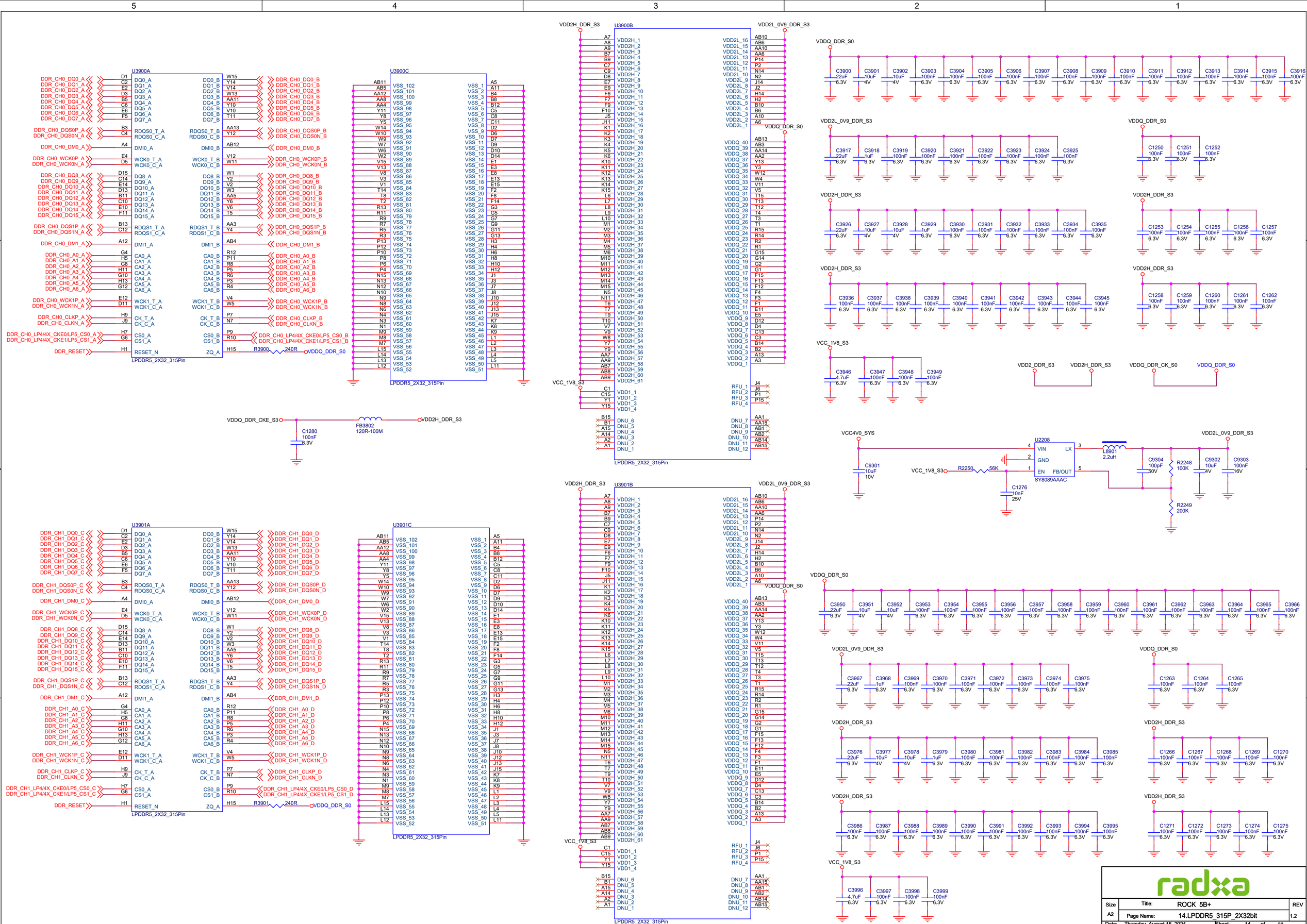
RK3588-Socket

VCC_3V3_S0

R95032 2.2K

R95033 2.2K

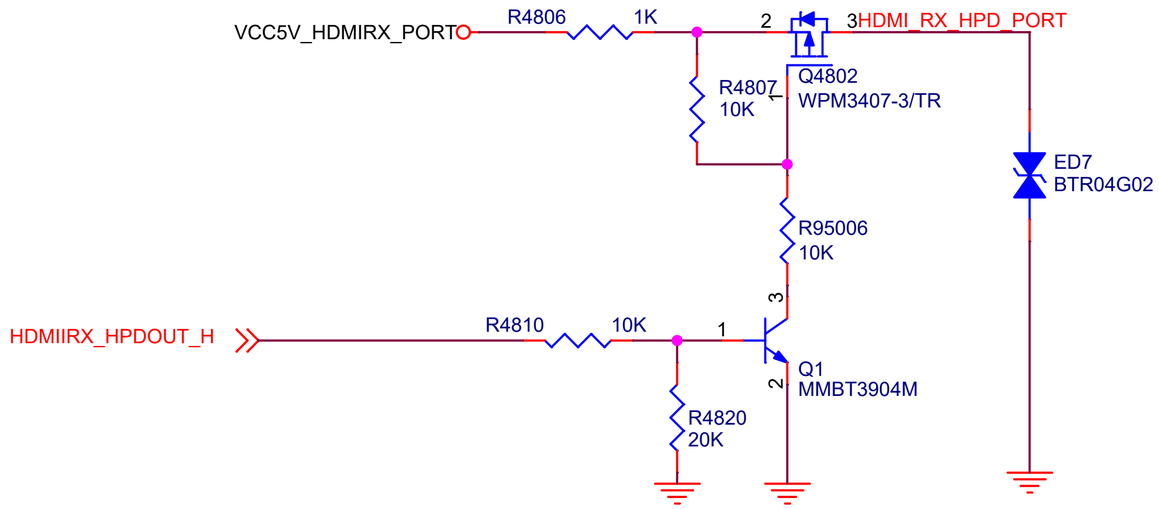
AK30		CM_RST_L
AL30		HDMI1_TX_ON_H
AM29		W-DISABLE
AL29		PCIE30x1_0_CLKREQn_M1_L
AL28		PCIE30x1_0_WAKEn_M1_L
AK27		PCIE30x1_0_PERSTn_M1_L
AL27		PCIE30x2_CLKREQn_M1
AM27		PCIE30x2_WAKEn_M1
AK26		PCIE30x2_PERSTn_M1
AL24		HDMI0_TX_ON_H
AK25		I2C7_SCL_M3
AM25		I2C7_SDA_M3
AL26		PCIE30X4_CLKREQn_M1_L
AJ26		PCIE30X4_WAKEn_M1_L
AJ27		PCIE30X4_PERSTn_M1_L
AJ28		HDMITX0_SCL_M0
AJ25		HDMITX0_SDA_M0
AK24		HDMITX0_CEC_M0

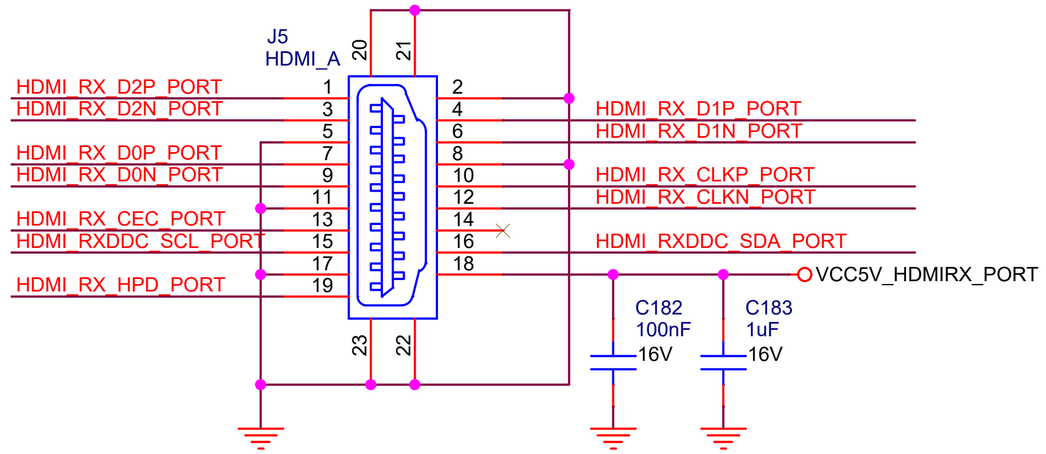


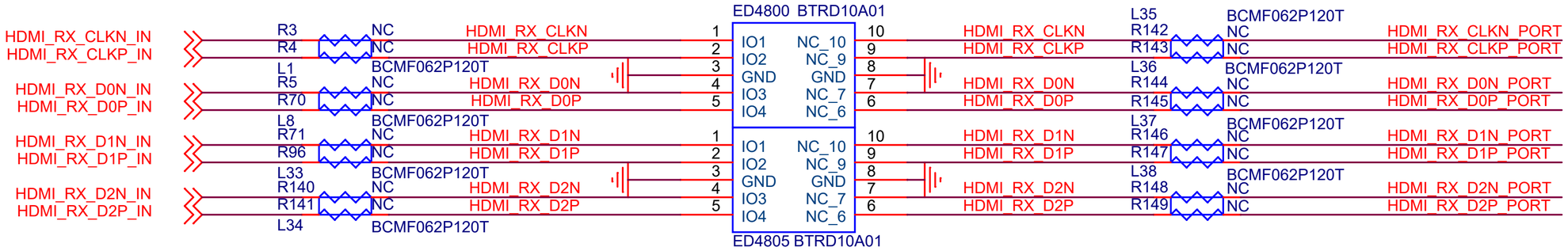
Size A2	Title: ROCK 5B+	REV 1.2
	Page Name: 14.LPDDR5_315P_2X32bit	
Date: Thursday, August 15, 2024	Sheet 14 of 33	

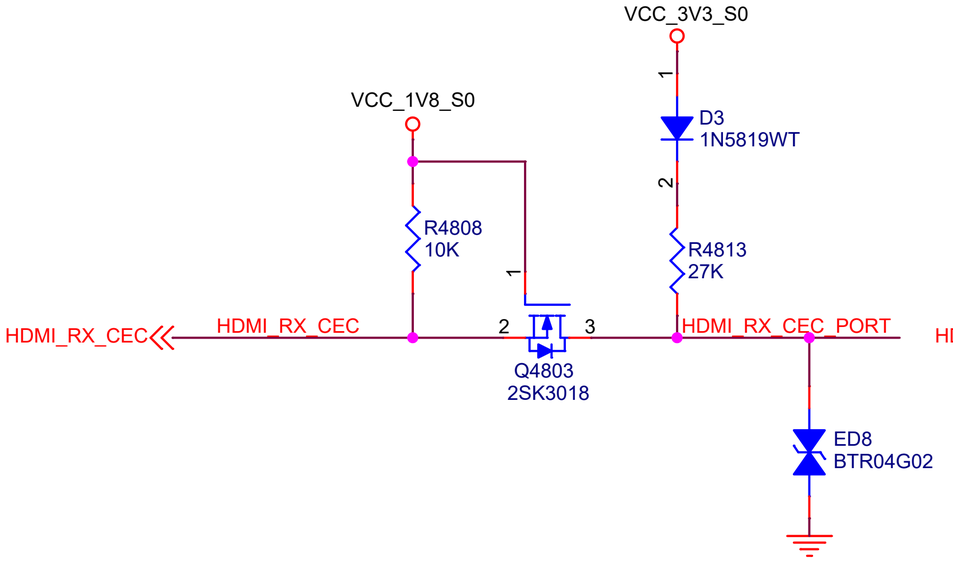
SDMMC_DET_L
R346
100K
VCC1V8_PMU_DDR_S3
VCC_3V3_S3
R13
FB120R
C546
10uF
10V
C547
100nF
10V
SDMMC_D2/JTAG_TCK_M0
SDMMC_D3/JTAG_TMS_M0
SDMMC_CMD/MCU_JTAG_TCK_M0
SDMMC_CLK/MCU_JTAG_TMS_M0
R12
0R
SDMMC0_D0
SDMMC0_D1
D9
D8
D7
C185
NC
D6
D5
D4
D11
BTR04G02
C
G
H
D
G4
G7
G8
G3
1
2
3
4
5
6
7
8
9
10
DATA2
CD/DATA3
CMD
VDD
CLK
VSS
DATA0
DATA1
D1
D2
J4
SMD_TF-008
G1
G5
G6
G2
A
E
F
B
VCC_3V3_S3
R9
10K
SDMMC_DET_L
3
1
R76
22R
2
Q5
MMBT3904M
R77
10K
SDMMC0_D0
SDMMC0_D1
SDMMC_D2/JTAG_TCK_M0
SDMMC_D3/JTAG_TMS_M0
SDMMC_CMD/MCU_JTAG_TCK_M0
SDMMC_CLK/MCU_JTAG_TMS_M0
SDMMC_DET_L

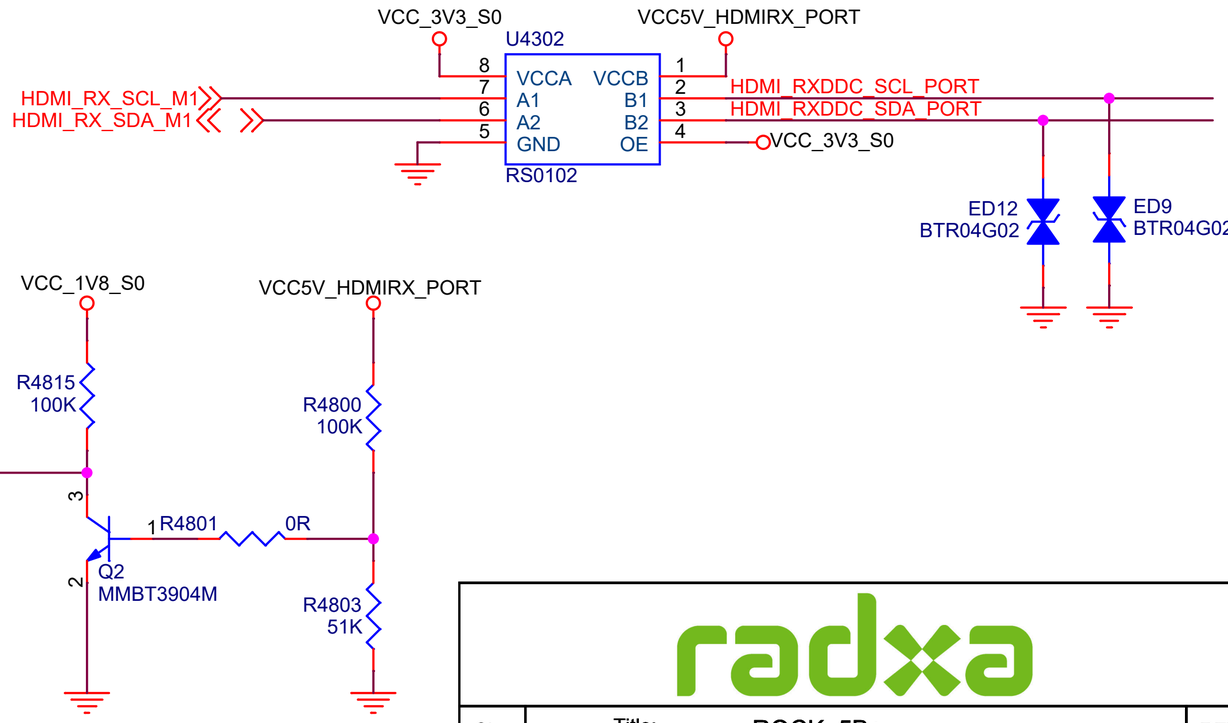
Size A4	Title: ROCK 5B+	REV 1.2
	Page Name: 15.Flash-TF Card	
Date: Thursday, August 15, 2024	Sheet 15 of 33	



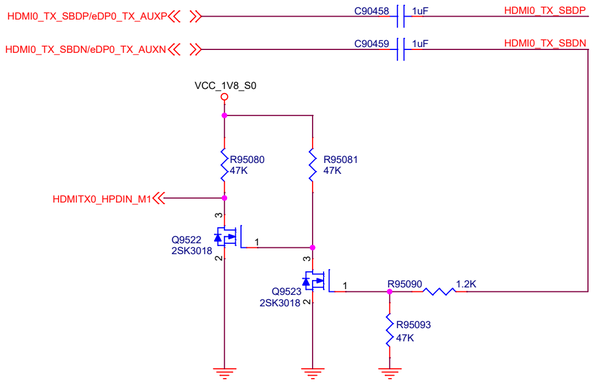


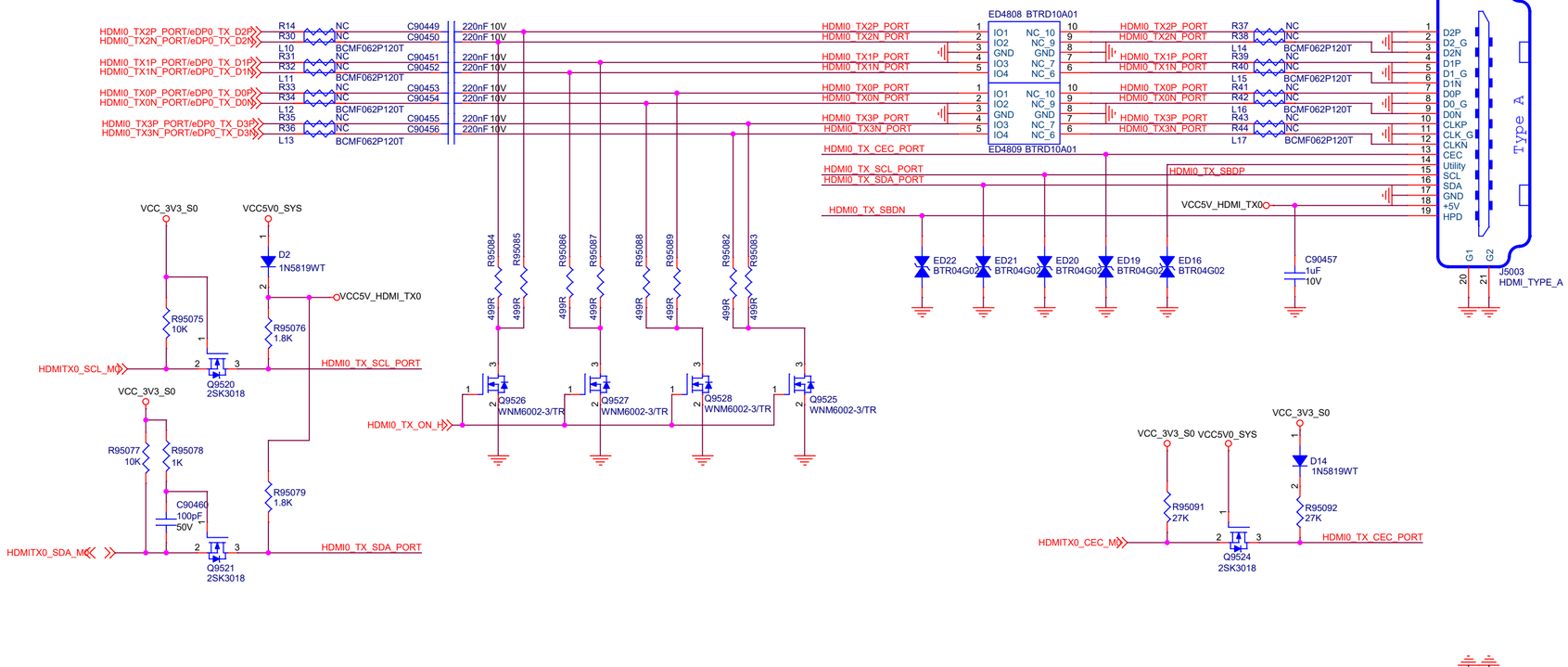


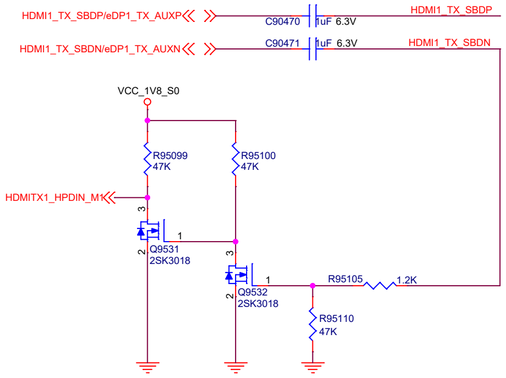


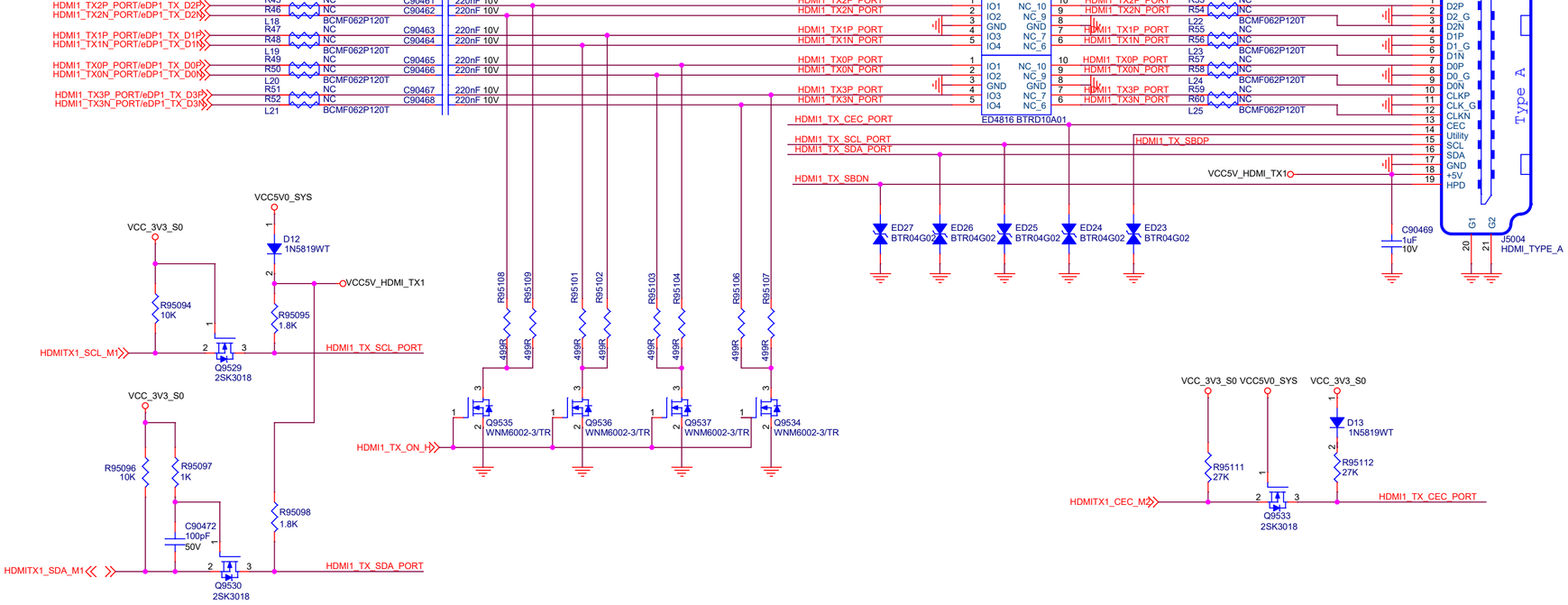


Size A4	Title: ROCK 5B+	REV 1.2
	Page Name: 16.VI-HDMI2.0 RX	
Date: Thursday, August 15, 2024	Sheet 16 of 33	

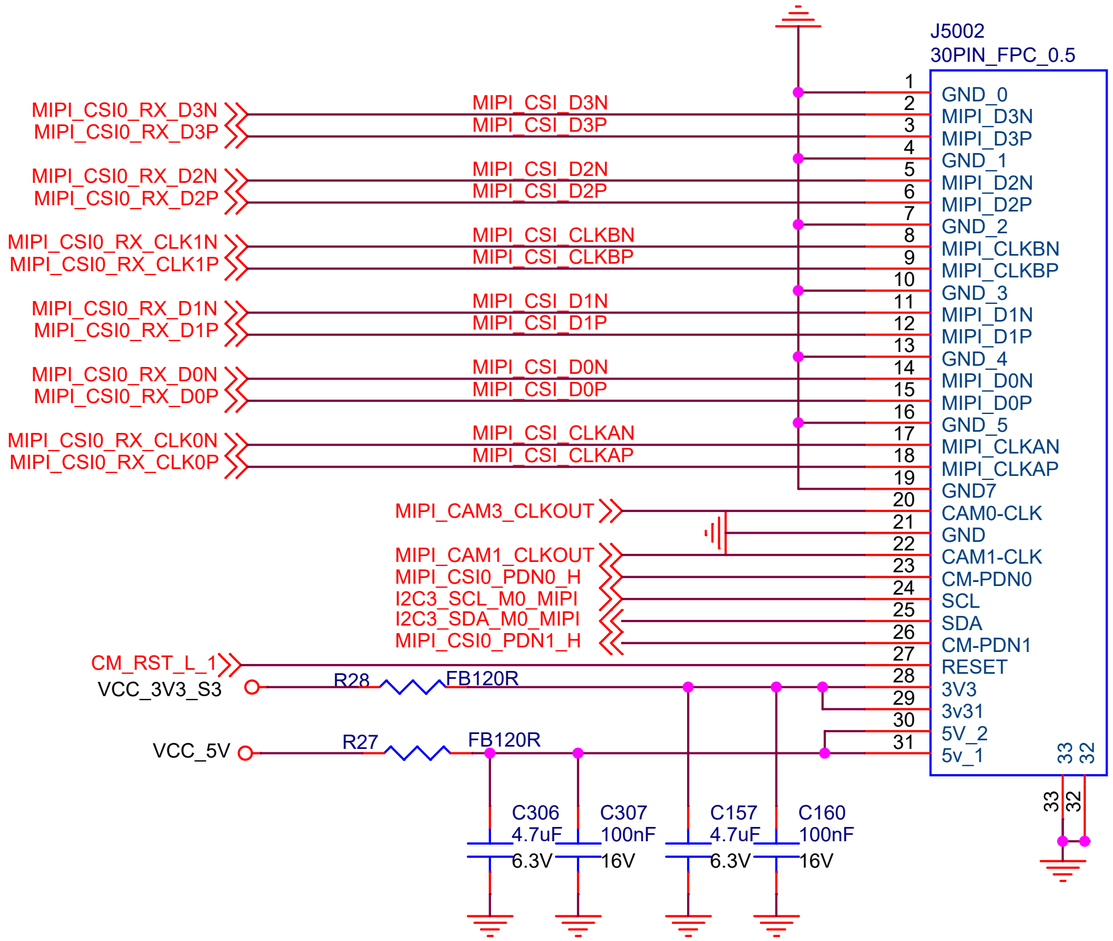


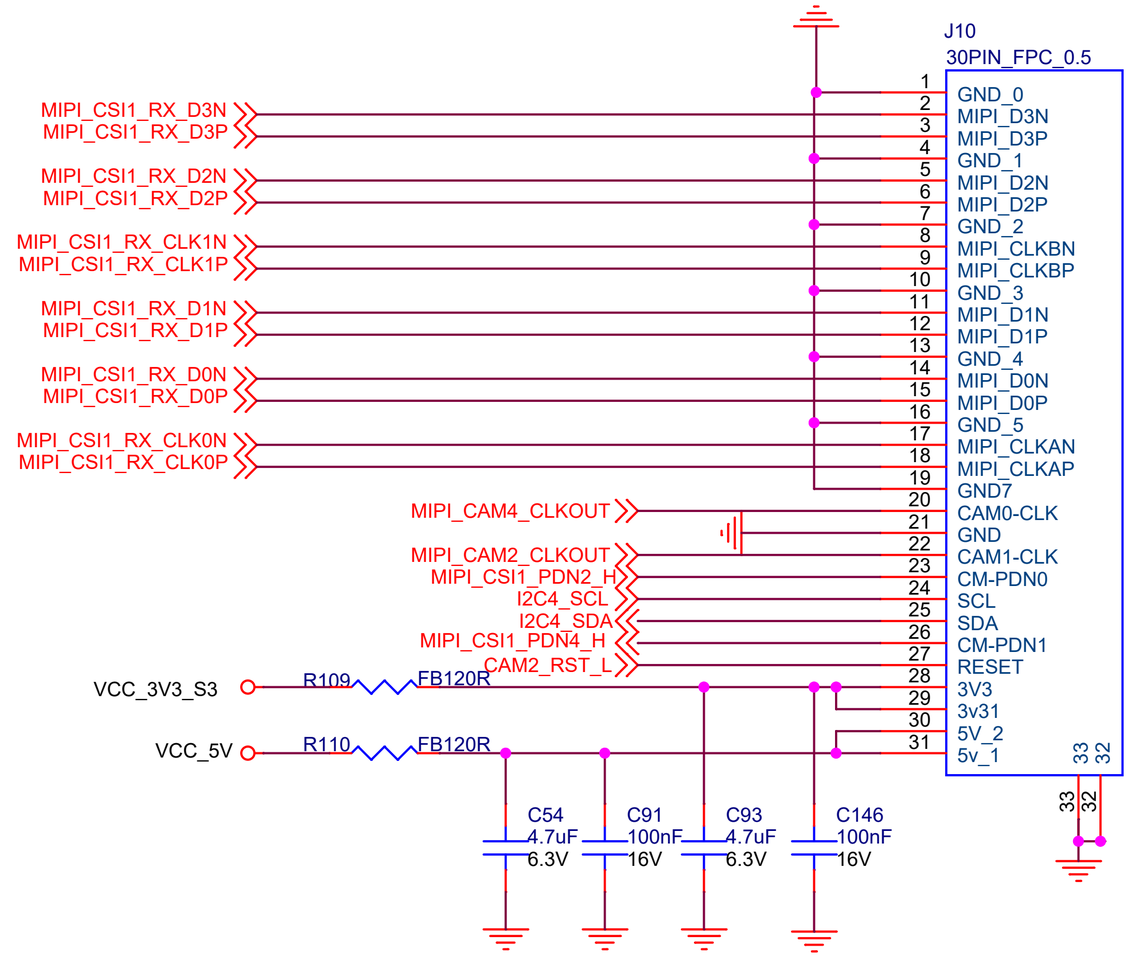


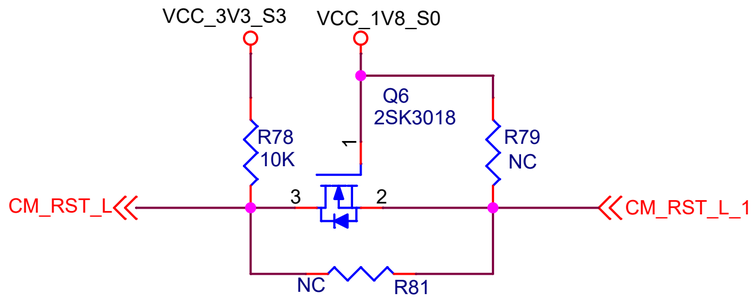




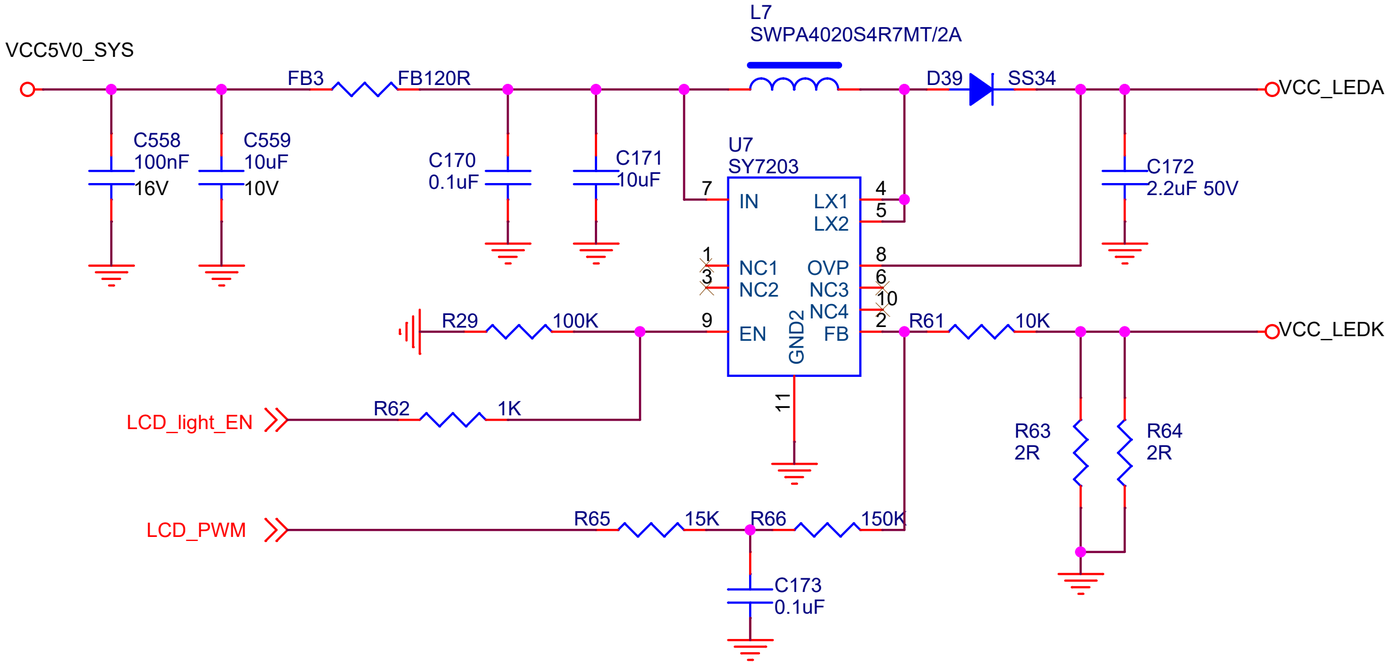
Size A2	Title: ROCK 5B+		REV 1.2
	Page Name: 17.VO-HDMI2.1 TX		
Date: Thursday, August 15, 2024	Sheet 17 of 33		

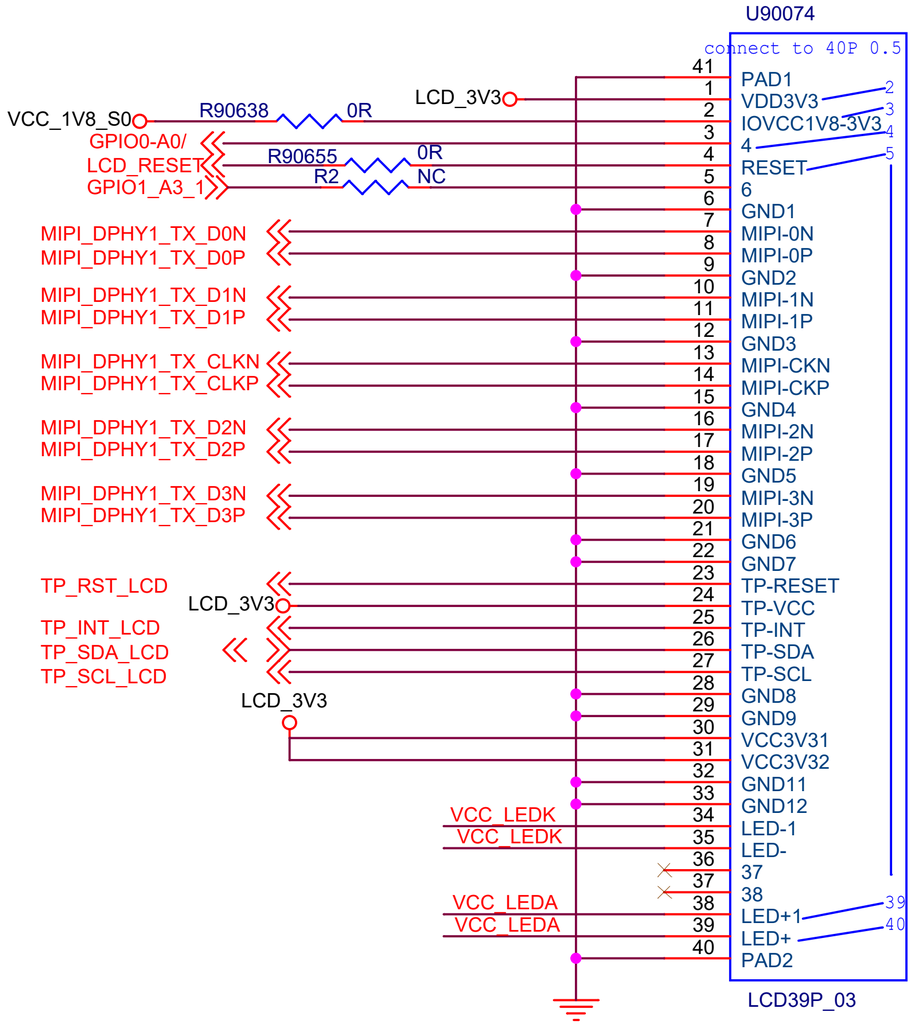


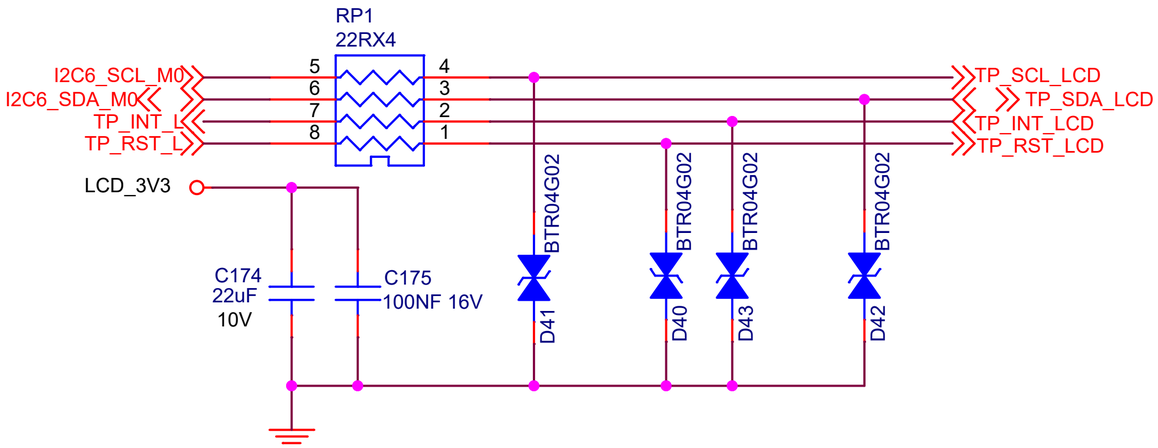




Size A4	Title: ROCK 5B+	REV 1.2
	Page Name: 18.Camera_MIPI-CSI	
Date: Thursday, August 15, 2024	Sheet 18 of 33	

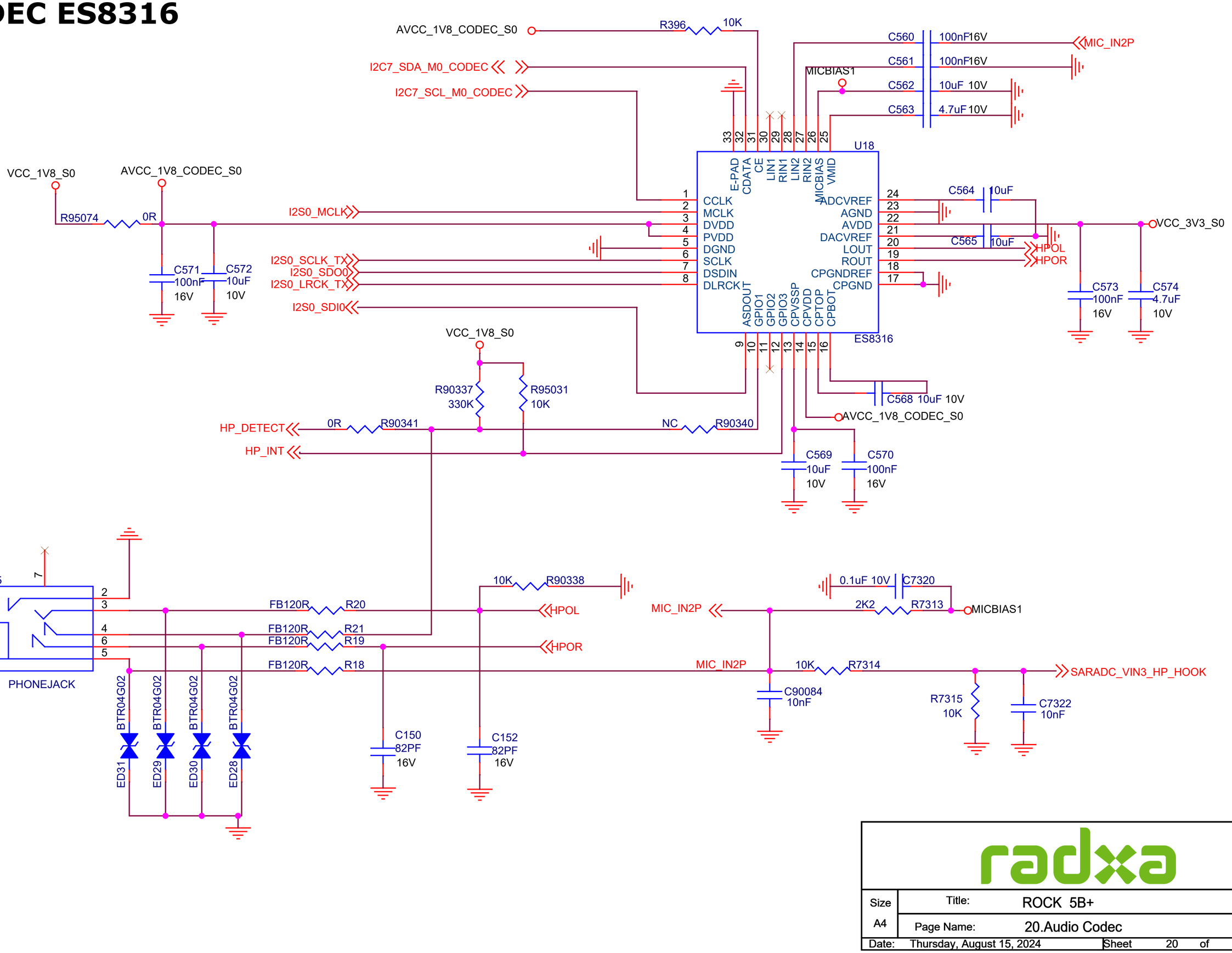






Size A4	Title: ROCK 5B+	REV 1.2
	Page Name: 19.VO-LCM_MIPI	
Date: Thursday, August 15, 2024	Sheet 19 of 33	

CODEC ES8316



Size A4	Title: ROCK 5B+	REV 1.2
	Page Name: 20.Audio Codec	
Date: Thursday, August 15, 2024	Sheet 20 of 33	

U13B
EMMC_B153_2L
A7 E5 E8 G3 G10 K6 K7 P7 P10
RFU1 RFU2 RFU3 RFU4 RFU5 RFU6 RFU7 RFU8 RFU9
A2 NC2
A8 NC8
A9 NC9
A10 NC10
A11 NC11
A12 NC12
A13 NC13
A14 NC14
B1 NC15
B7 NC21
B8 NC22
B9 NC23
B10 NC24
B11 NC25
B12 NC26
B13 NC27
B14 NC28
C1 NC29
C3 NC31
C7 NC35
C8 NC36
C9 NC37
C10 NC38
C11 NC39
C12 NC40
C13 NC41
C14 NC42
D1 NC43
D2 NC44
D3 NC45
D4 NC46
D12 NC54
D13 NC55
D14 NC56
E1 NC57
E2 NC58
E3 NC59
E12 NC68
E13 NC69
E14 NC70
F1 NC71
F2 NC72
F3 NC73
F12 NC82
F13 NC83
F14 NC84
G1 NC85
G2 NC86
G12 NC96
G13 NC97
G14 NC98
NC196 P14
NC195 P13
NC194 P12
NC193 P11
NC191 P9
NC190 P8
NC184 P2
NC183 P1
NC182 N14
NC181 N13
NC180 N12
NC179 N11
NC178 N10
NC177 N9
NC176 N8
NC175 N7
NC174 N6
NC171 N3
NC169 N1
NC168 M14
NC167 M13
NC166 M12
NC165 M11
NC164 M10
NC163 M9
NC162 M8
NC161 M7
NC157 M3
NC156 M2
NC155 M1
NC154 L14
NC153 L13
NC152 L12
NC143 L3
NC142 L2
NC141 L1
NC140 K14
NC139 K13
NC138 K12
NC129 K3
NC128 K2
NC127 K1
NC126 J14
NC125 J13
NC124 J12
NC115 J3
NC114 J2
NC113 J1
NC112 H14
NC111 H13
NC110 H12
NC101 H3
NC100 H2
NC99 H1

U13A
EMMC_B153_2L
eMMC_D0 A3 DATA0
eMMC_D1 A4 DATA1
eMMC_D2 A5 DATA2
eMMC_D3 B2 DATA3
eMMC_D4 B3 DATA4
eMMC_D5 B4 DATA5
eMMC_D6 B5 DATA6
eMMC_D7 B6 DATA7
eMMC_CMD M5 CMD
eMMC_CLKOUT M6 CLK
eMMC_RSTn K5 RST_n
eMMC_DATA_STROBE H5 Data Strobe
C193 10pF 25V
R103 10K 5%
VCC_1V8_S3
C199 2.2uF 6.3V C2 VDDi
R138 22R
R139 47K
E9 VSF1
E10 VSF2
F10 VSF3
K10 VSF4
VCCQ1 C6
VCCQ2 M4
VCCQ3 N4
VCCQ4 P3
VCCQ5 P5
VCC1 E6
VCC2 F5
VCC3 J10
VCC4 K9
VSS1 J5
VSS2 A6
VSS3 C4
VSS4 E7
VSS5 G5
VSS6 H10
VSSQ1 K8
VSSQ2 N2
VSSQ3 N5
VSSQ4 P4
VSSQ5 P6
VCC_1V8_S3
C186 100nF 16V
C187 100nF 16V
C190 100nF 16V
C191 100nF 16V
C192 4.7uF 10V
VCC_3V3_S3
C194 100nF 16V
C197 100nF 16V
C198 4.7uF 10V

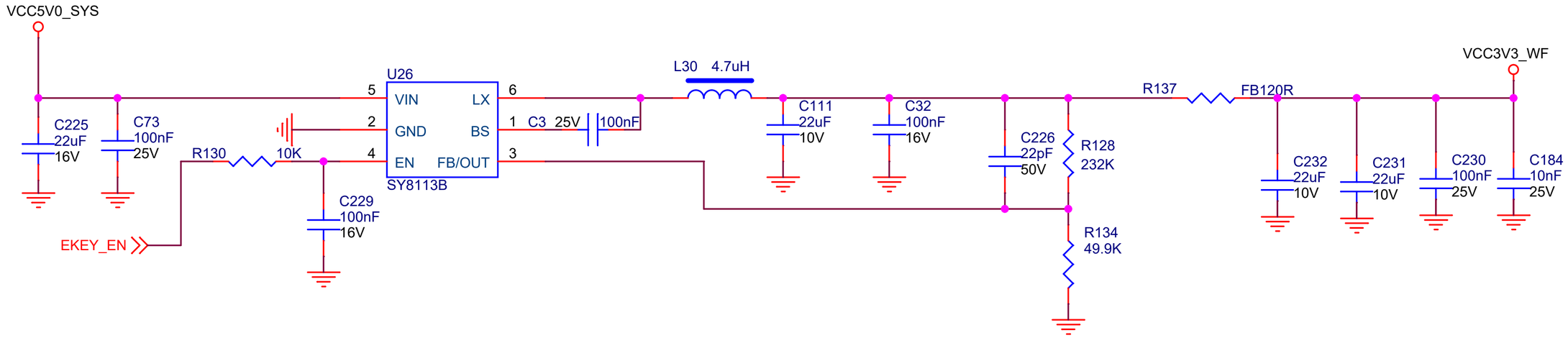
VCC_1V8_S3
R99 10K eMMC_D0
R100 10K eMMC_CMD

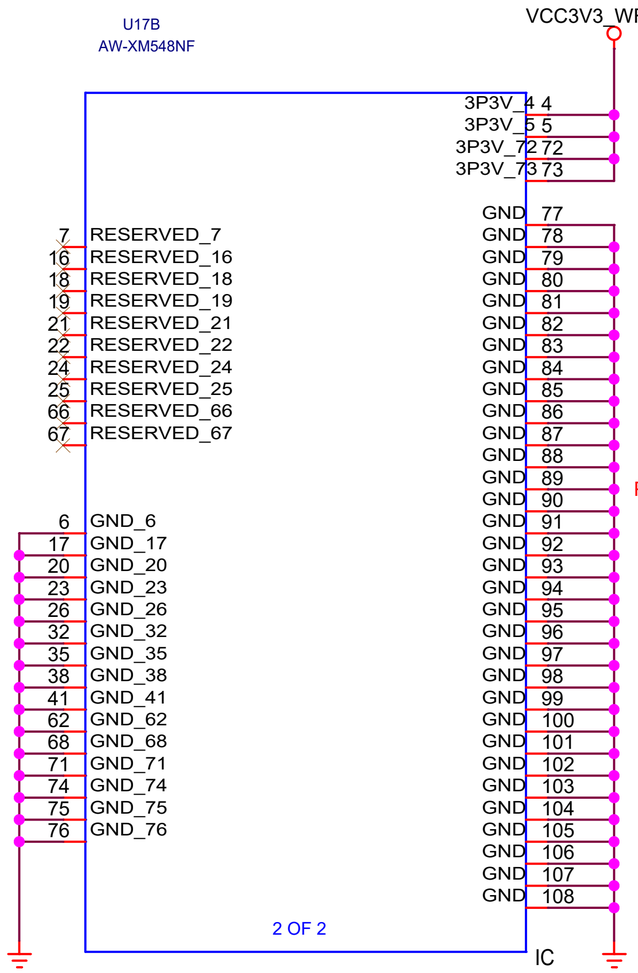
J3
2P_2.0
VCC1V8_PMU_DDR_S3
FSPI-CLK
R164 22R
MMBT3904M
Q7
R167 10K
R169 NC
ED6
BTR04G02

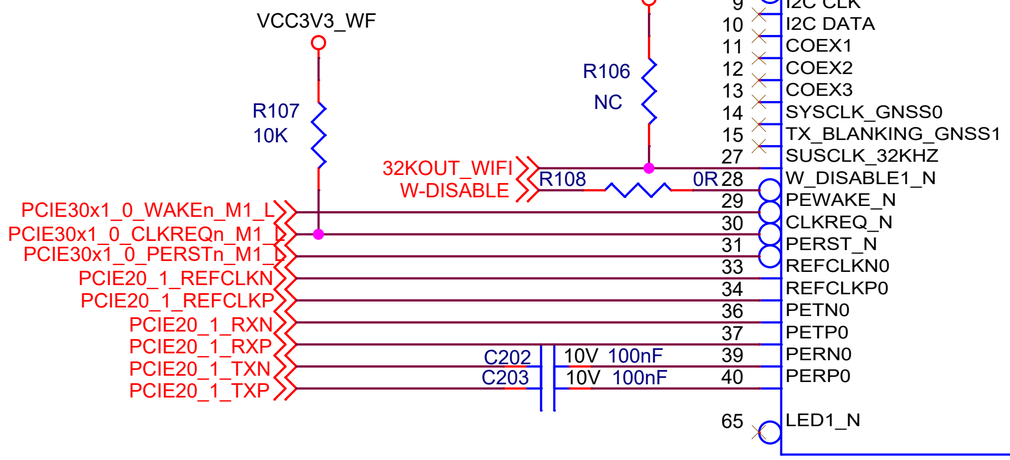
VCC_3V3_S3
R4300 10K
R4301 10K
U4300
XT25F128BWOI/HGA
FSPI-CS0 1 CS
FSPI-D1 2 DO(D1)
FSPI-D2 3 WP(D2)
4 VSS
EPAD 9
8 VCC
7 HOLD(D3)
6 CLK
5 DI(D0)
FSPI-D3
FSPI-CLK
FSPI-D0
C4300 100nF 16V
C4301 1uF 16V

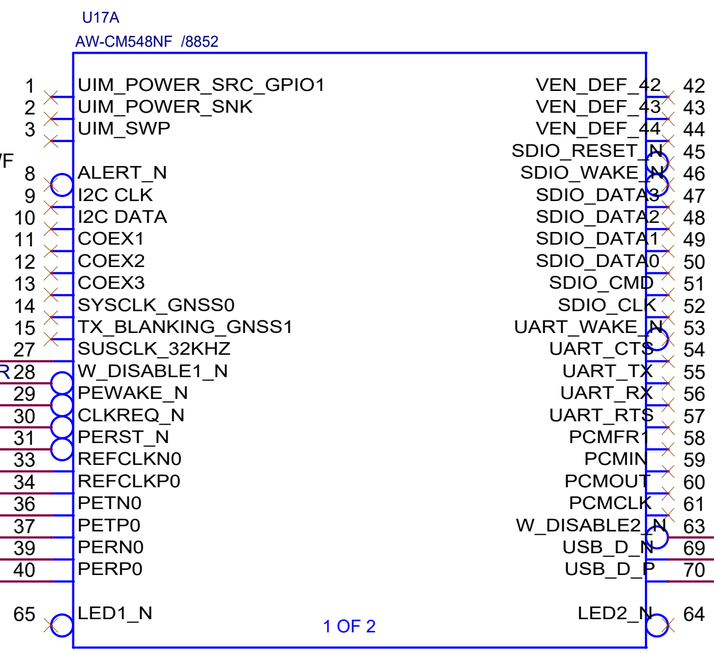
radxa

Size	Title:	REV
A4	ROCK 5B+	
	Page Name: 21.SPI/eMMC Flash	1.2
Date: Thursday, August 15, 2024	Sheet 21 of 33	









Size	Title: ROCK 5B+	REV
A4	Page Name: 22.PCIE_WIFI	1.2
Date: Thursday, August 15, 2024	Sheet 22 of 33	

U12
M2-KEY-B

	1	CONFIG_3			
	3	GND13	VBAT1	2	3V3_4G
	5	GND14	VBAT2	4	
USB4_Dp R104 0R	7	USB_D+	FULL_CARD_PWR_OFF#	6	4G_PWREN
USB4_DM R105 0R	9	USB_D-	W_DISABLE1#	8	4G_DISABLE
	11	GND20	LED1#	10	

PCIE M.2 NGFF B-KEY SOCKET

	21	CONFIG_0	I2S_CLK	20	R113 0R I2S2_LRCK_M0_BT
4G_WAKE_ON_HOST	23	WoWWAN#	I2S_TXD	22	R112 0R I2S2_SDI_M0_BT
	25	DPR	I2S_RXD	24	R111 0R I2S2_SDO_M0_BT
	27	GND9	W_DISABLE2#	26	W_DISABLE2
	29	U3_TX-	I2S_WA	28	
	31	U3_TX+	SIM1_RESET	30	SIM1_RESET
	33	GND8	SIM1_CLK	32	SIM1_CLK
	35	U3_RX-	SIM1_DATA	34	SIM1_DATA
	37	U3_RX+	SIM1_PWR	36	SIM1_PWR
	39	GND7	I2C_SCL	38	DEVSLP
	41	PERn0/SATA-B+	SIM2_DET	40	
	43	PERp0/SATA-B-	SIM2_DATA	42	
	45	GND6	SIM2_CLK	44	
	47	PETn0/SATA-A-	SIM2_RESET	46	
	49	PETp0/SATA-A+	SIM2_PWR	48	
	51	GND5	PERST#	50	
	53	REFCLKN	CLKREQ#	52	
	55	REFCLKP	PEWAKE#	54	
	57	GND4	RFFE_SCLK	56	
	59	ANTCTL0	RFFE_SDATA	58	
	61	ANTCTL1	I2S_MCLK	60	R114 0R I2S2_SCLK_M0_BT
	63	LAA_TX_EN	COEX2	62	
	65	WL_TX_EN	COEX1	64	
1.8v 4G_RESET	67	RESET#	SIM1_DET	66	SIM1_DET
	69	CONFIG_1	I2C_SDA	68	
	71	GND3	VBAT3	70	
	73	GND2	VBAT4	72	
	75	CONFIG_2	VBAT5	74	3V3_4G
	80	79	hole	GND16	GND15
	80	79	78	77	76

C189 10uF 10V

SIM1_PWR
R84 15K

SIM1_CLK	R87	0R	SIM-CLK	3	CLK
SIM1_DATA	R90	0R	SIM-DATA	7	IO
SIM1_RESET	R91	0R	SIM-RESET	2	RST
				6	VPP
SIM1_PWR	R92	0R	SIM-PWR	1	VCC
SIM1_DET	R93	0R		5	GND
				SW	SW

U14
SMN-305-ARP7
13 12 11

BTR04G02 ED5
BTR04G02 ED4
BTR04G02 ED3
BTR04G02 ED10

C188 4.7uF 10V

VCC5V0_SYS

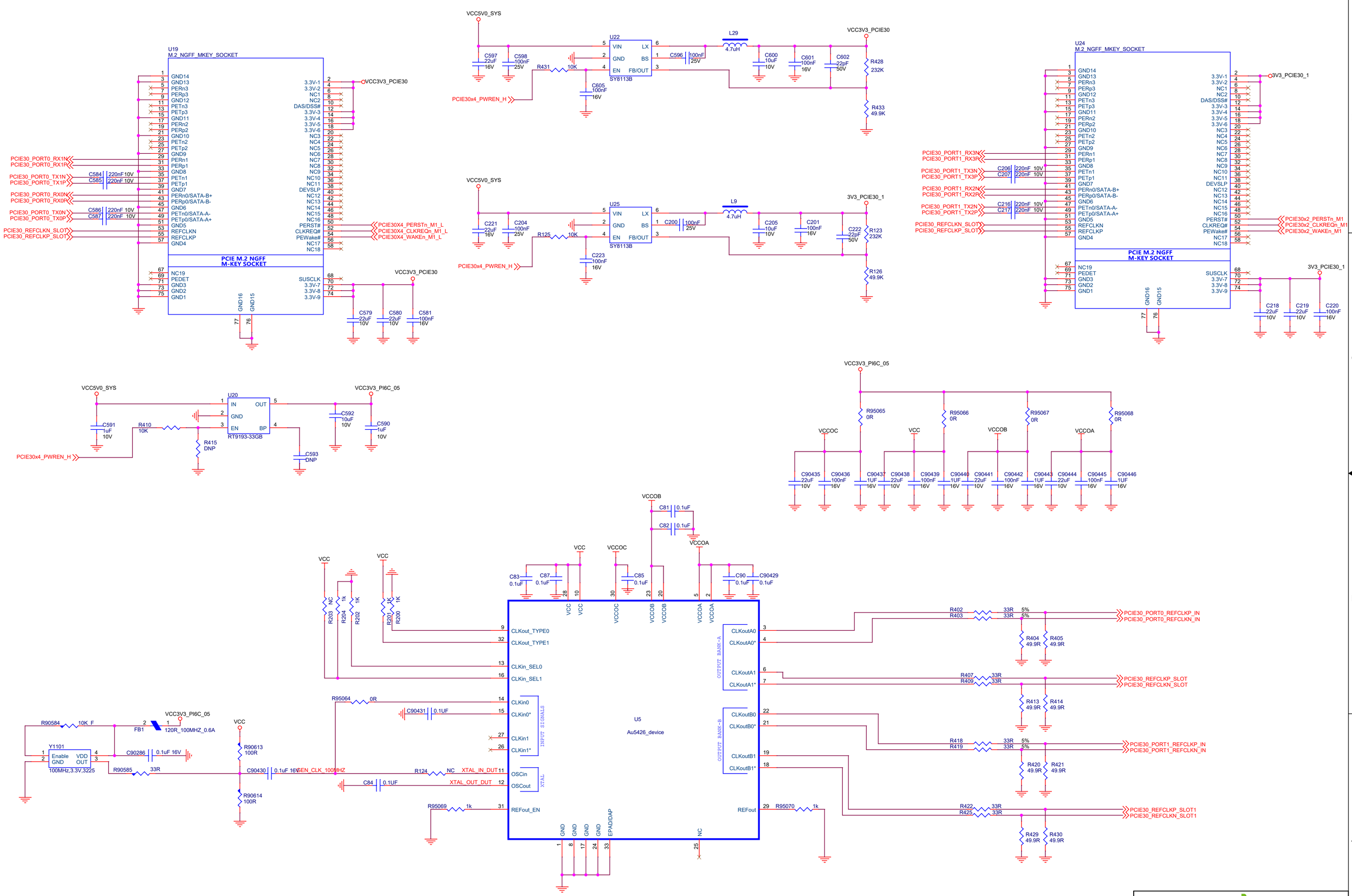
C209 22uF 10V
C210 100nF 25V
R136 NC
EKEY_EN
1K R132
C215 100nF 10V

U23
SY8113B/SM8103ADC

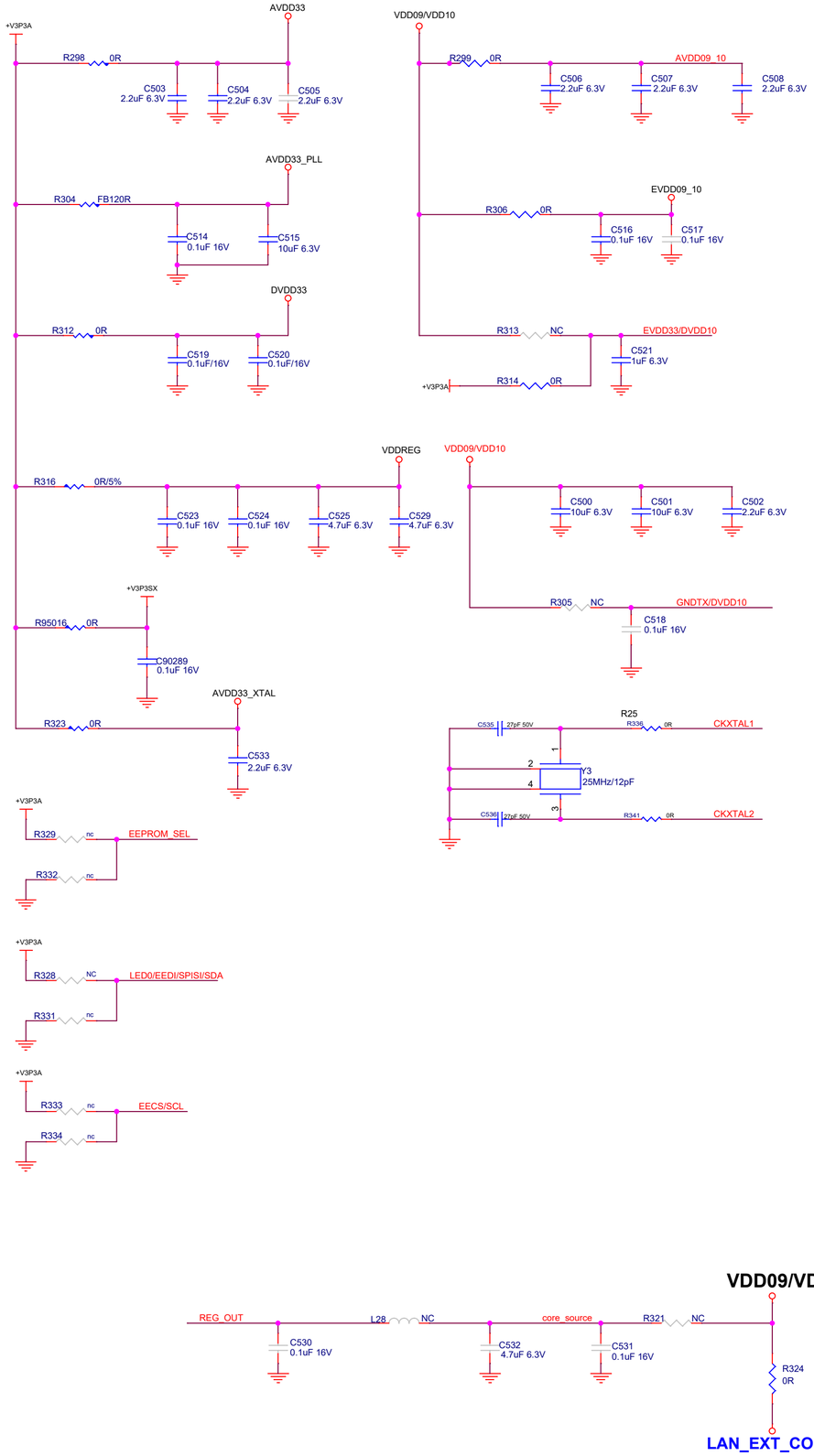
5	VIN	LX	6
2	GND	BS	1
4	EN	FB/OUT	3

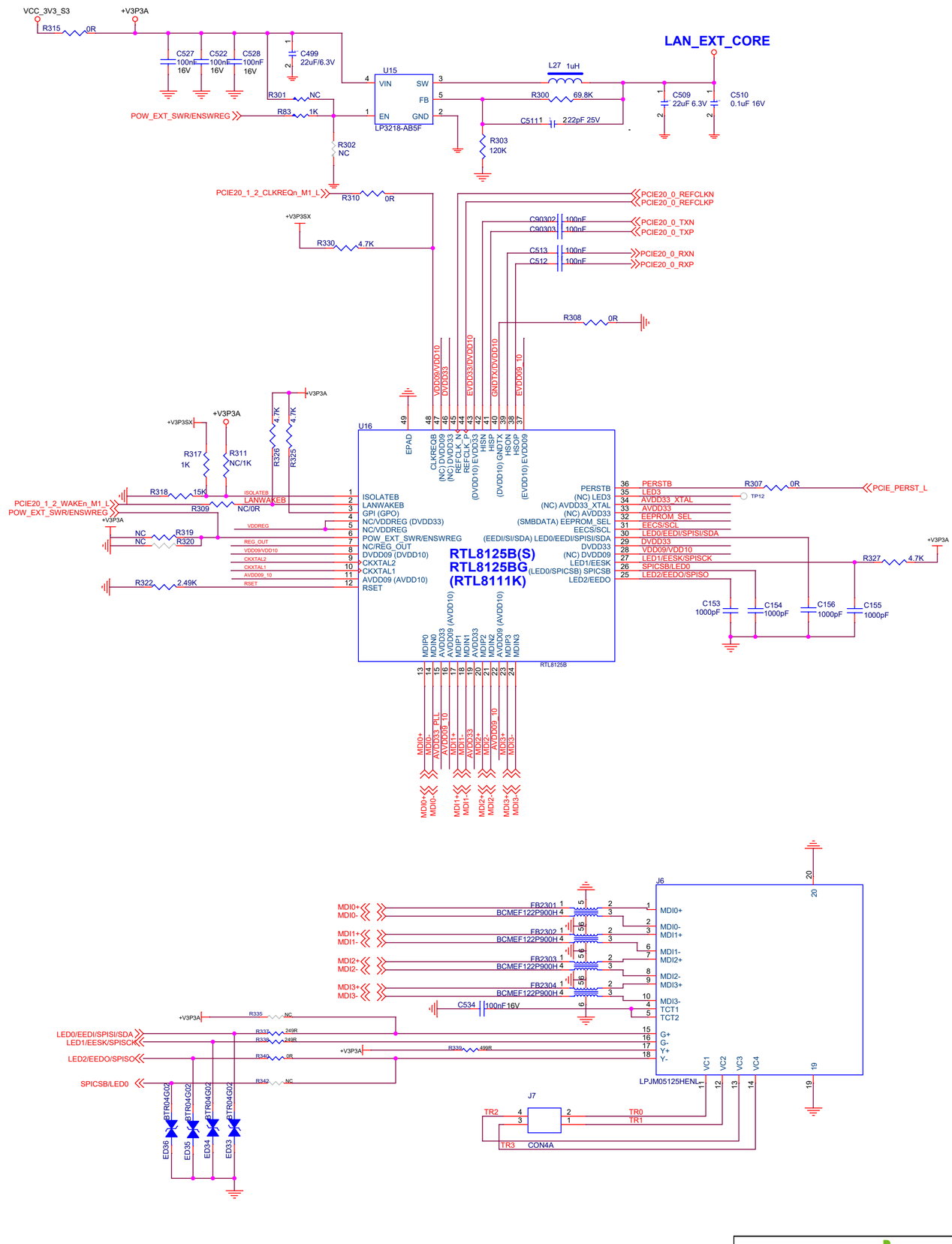
C208 100nF 25V
L26 RXWHJ3015-2R2MT
C212 22pF 50V
R133 232K
R135 49.9K
C211 22uF 10V
C213 22uF 10V
C214 100nF 10V
3V3_4G

Size	Title:	ROCK 5B+	REV
A4	Page Name:	23.M.2 B key_4G	1.2
Date:	Thursday, August 15, 2024	Sheet 23 of 33	

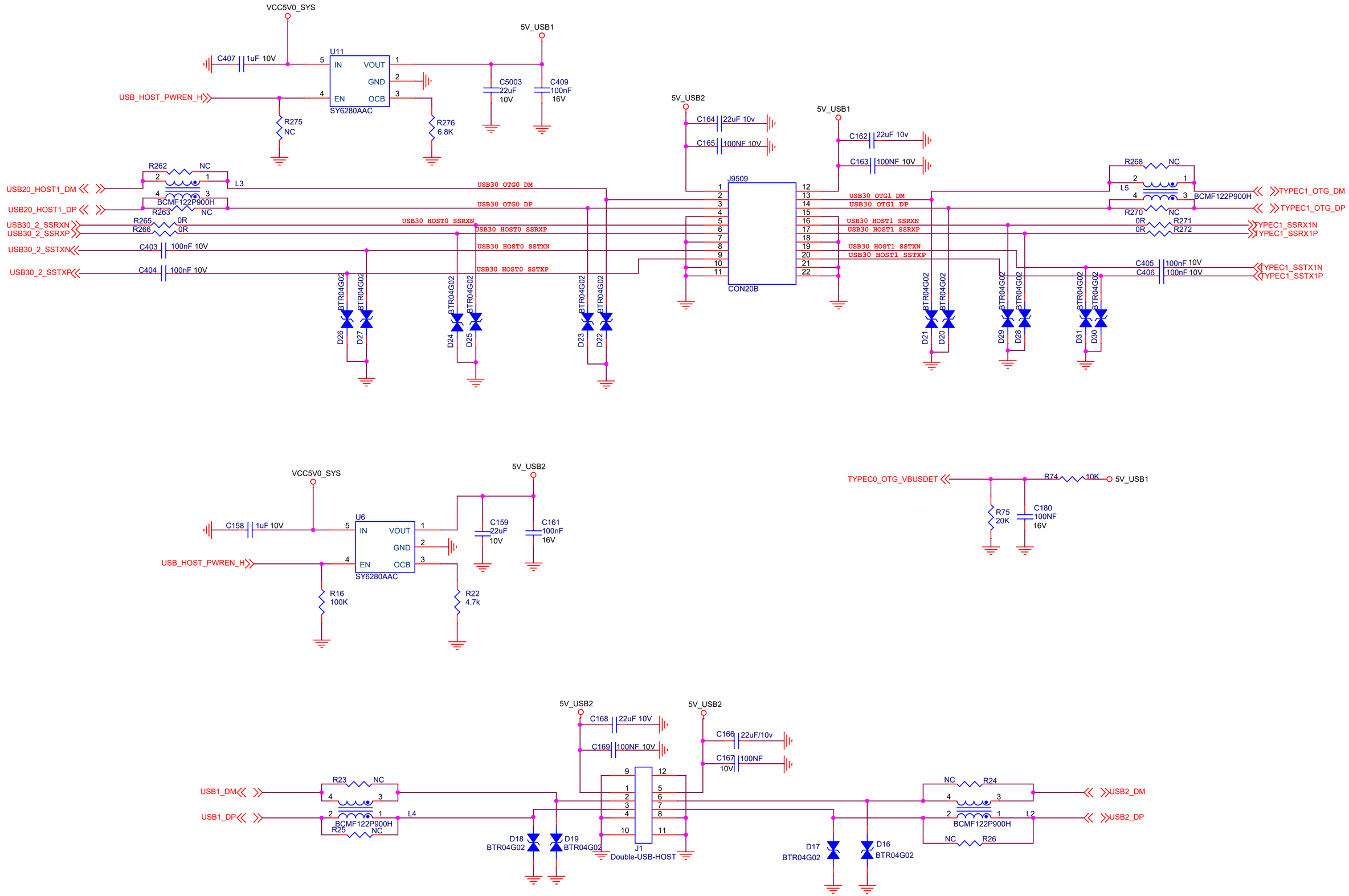


Size	Title:	ROCK 5B+	REV
A2	Page Name:	24.PCIE3.0 M key X2	1.2
Date:	Thursday, August 15, 2024	Sheet 24 of 33	

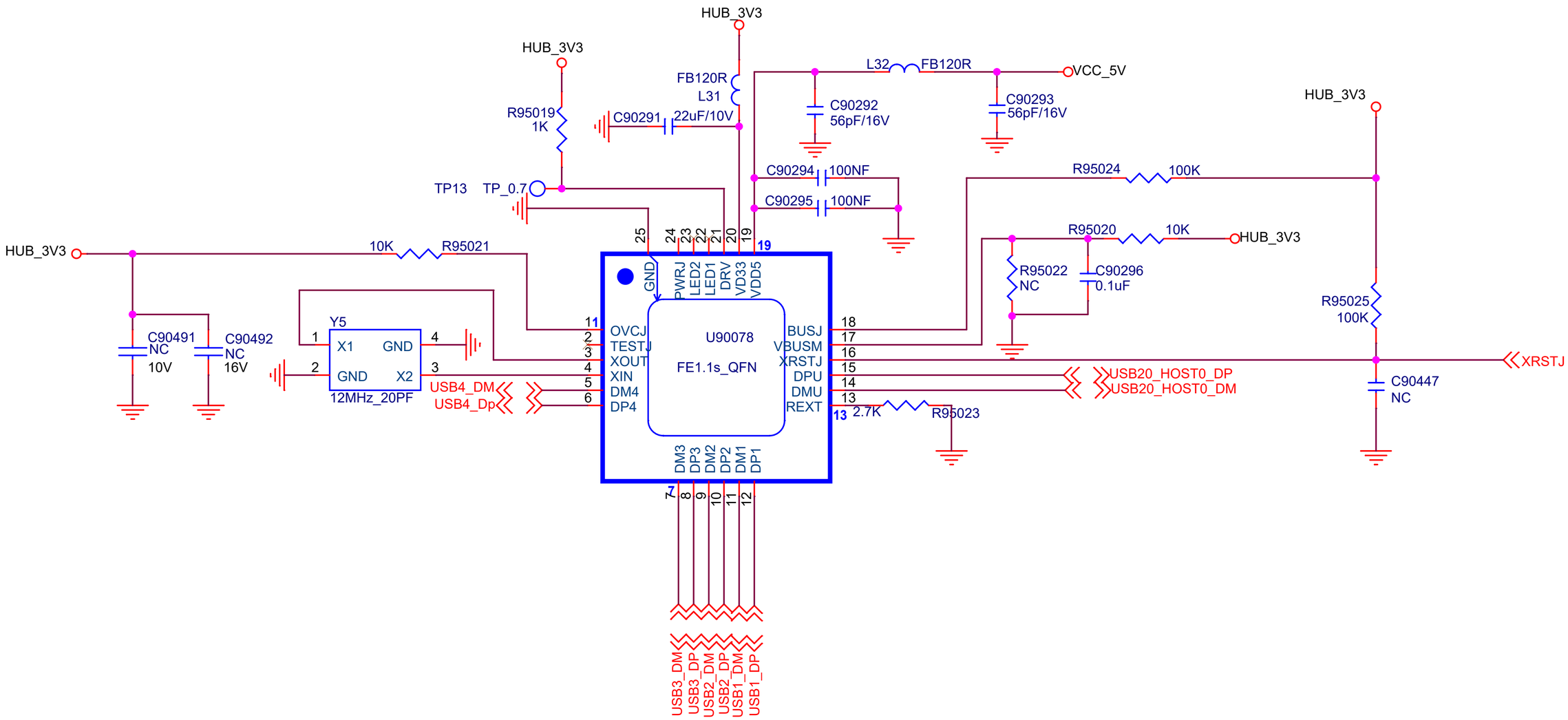




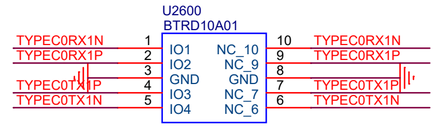
Size	Title:	ROCK 5B+	REV
A2	Page Name:	PCIE TO RJ45_RTL8125B	1.2
Date:	Thursday, August 15, 2024	Sheet 25 of 33	

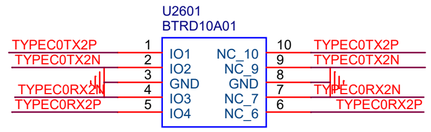


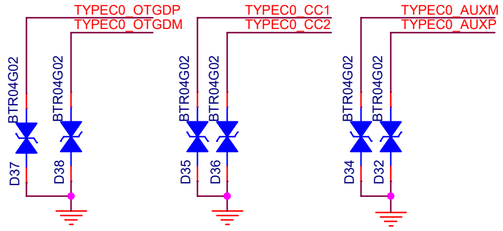
Size	Title: ROCK 5B+		REV
Custom	Page Name: USB3.0 HOST PORT		1.2
Date: Thursday, August 15, 2024	Sheet 26 of 33		

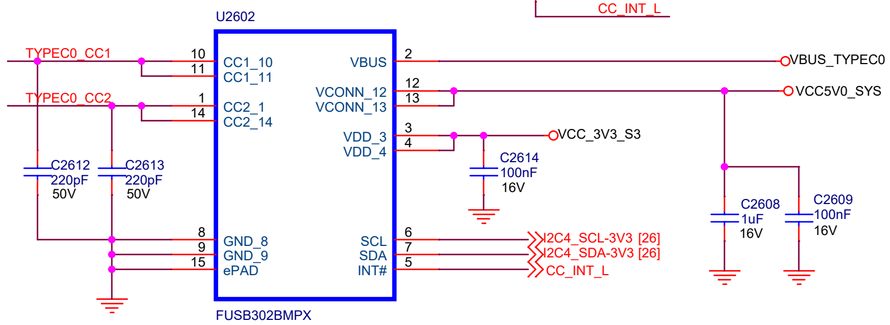


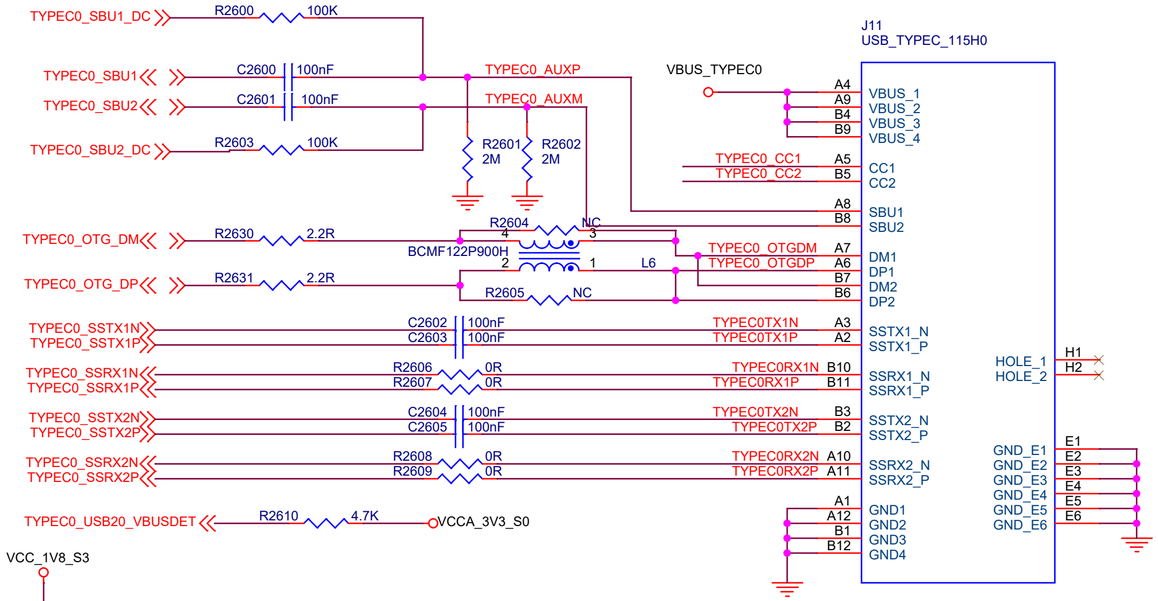
Size A4	Title: ROCK 5B+	REV 1.2
	Page Name: USB2HUB	
Date: Thursday, August 15, 2024	Sheet 27 of 33	

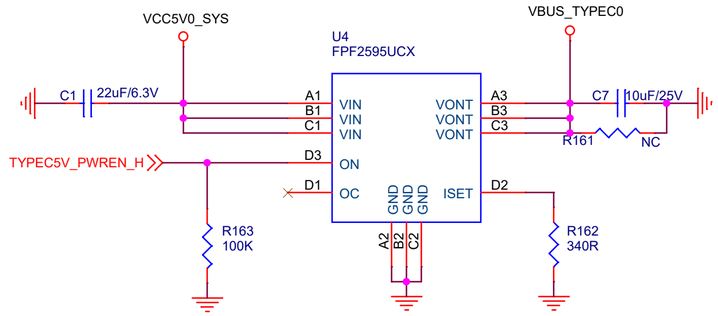


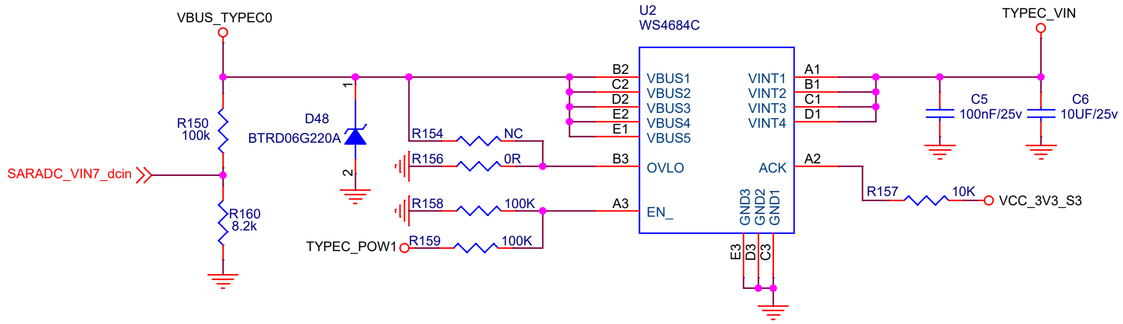




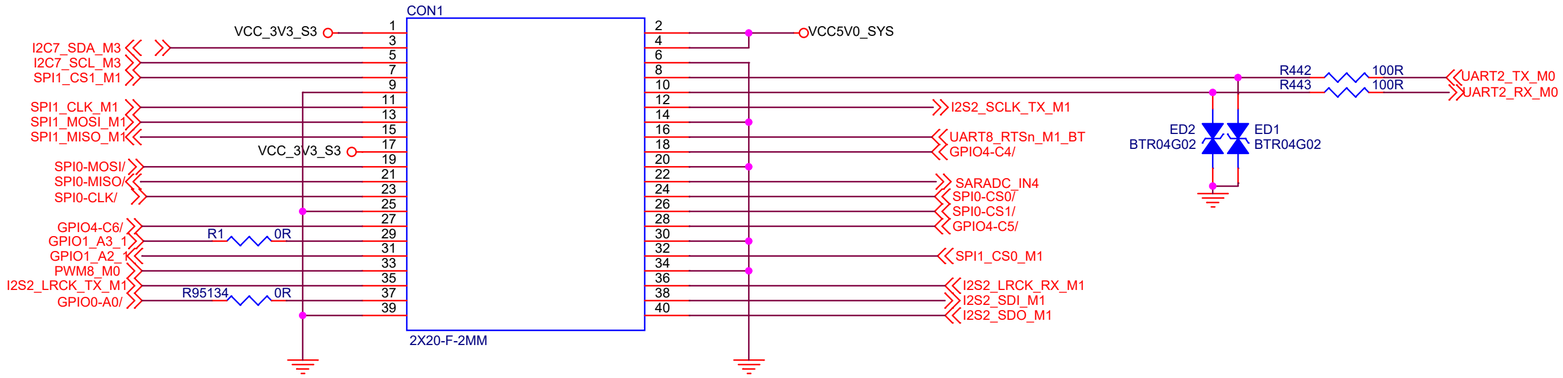


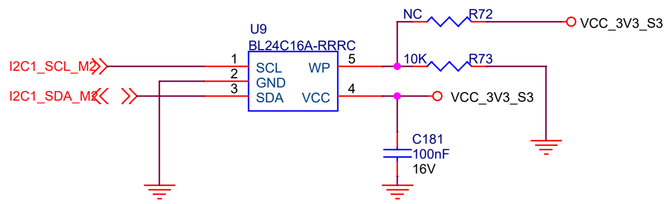


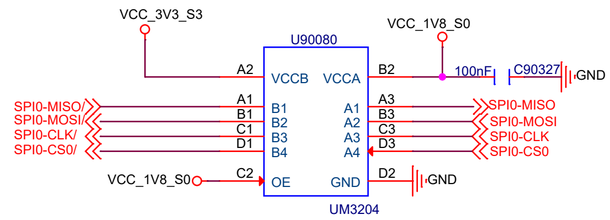


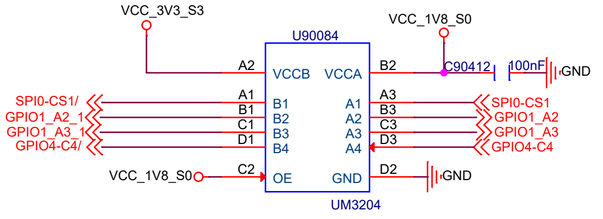


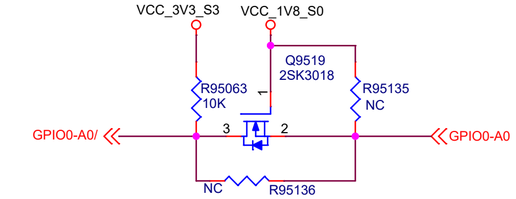
Size: Custom	Title: ROCK 5B+		REV: 1.2
	Page Name: Type-C PORT		
Date: Thursday, August 15, 2024	Sheet 28 of 33		



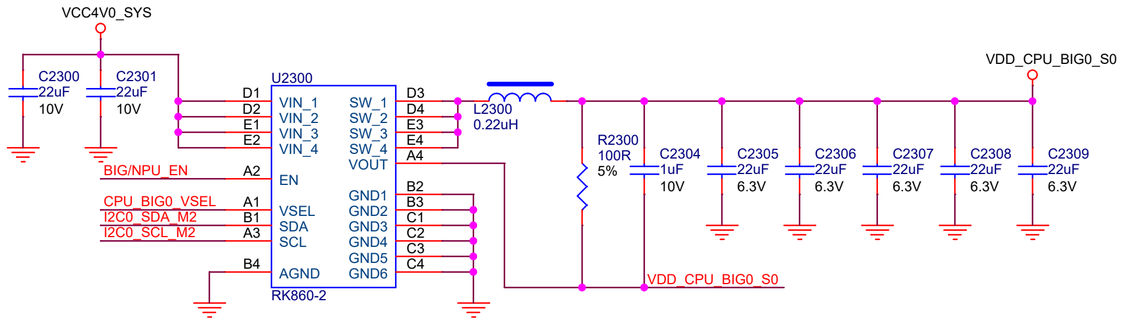


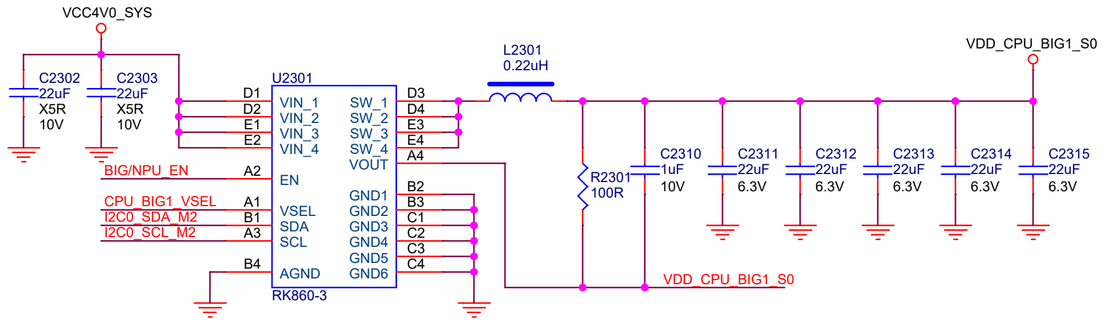


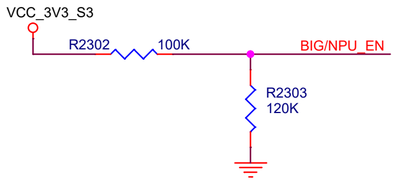


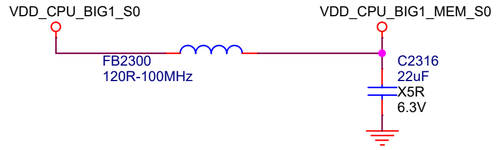


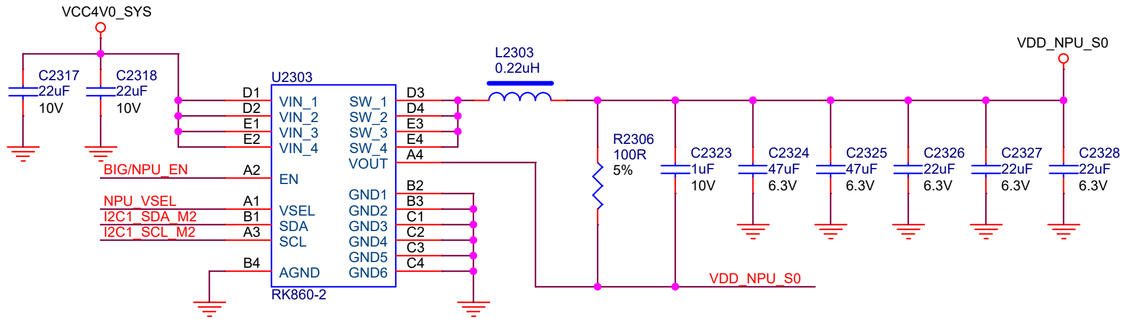
Size: Custom	Title: ROCK 5B+		REV: 1.2
	Page Name: CONNECT		
Date: Thursday, August 15, 2024		Sheet 29 of 33	

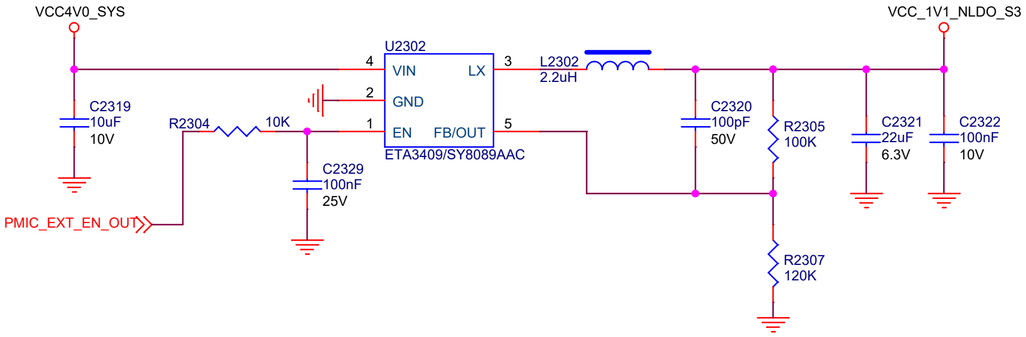


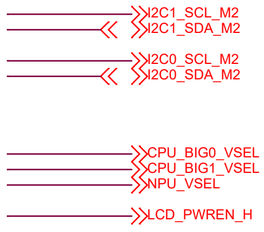


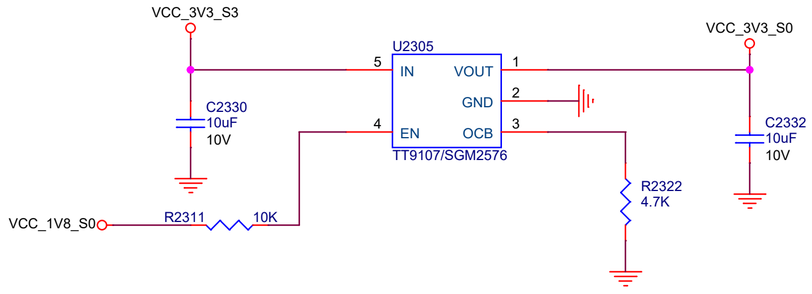


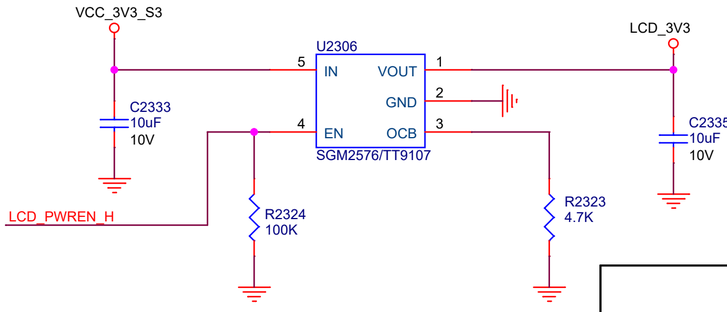




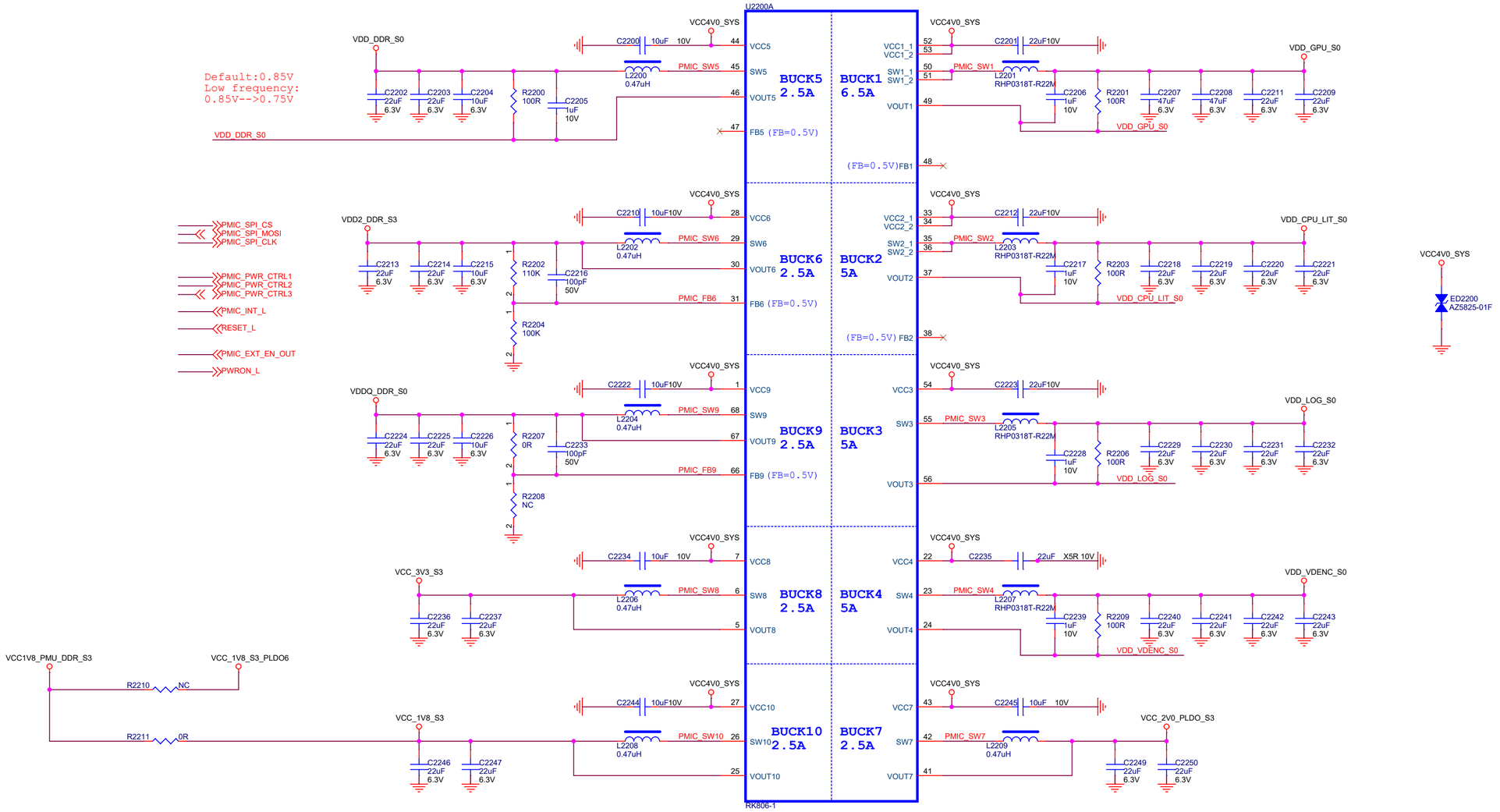


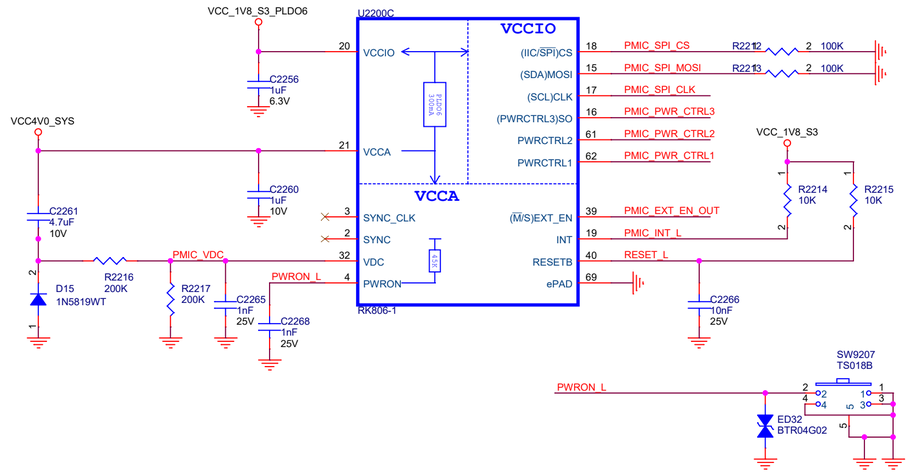


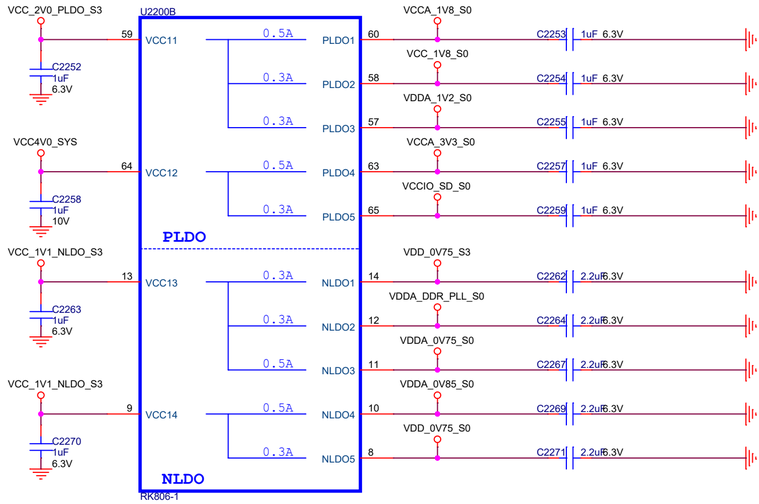




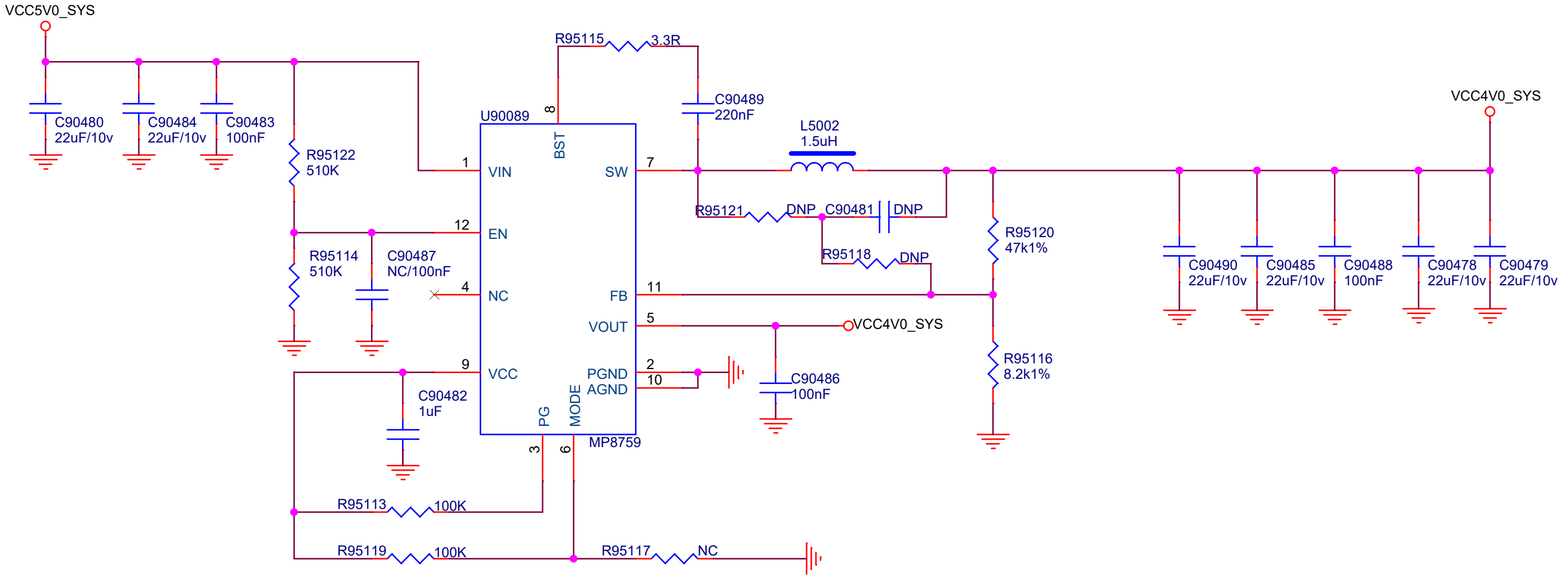
Size A3	Title: ROCK 5B+	REV 1.2
	Page Name: Power_Ext Discrete	
Date: Thursday, August 15, 2024	Sheet 30 of 33	

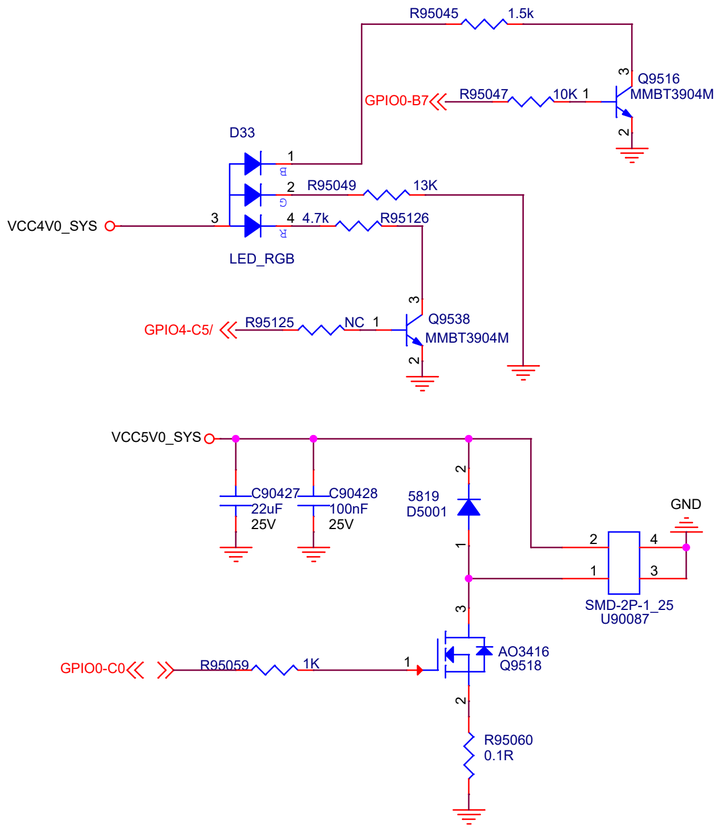


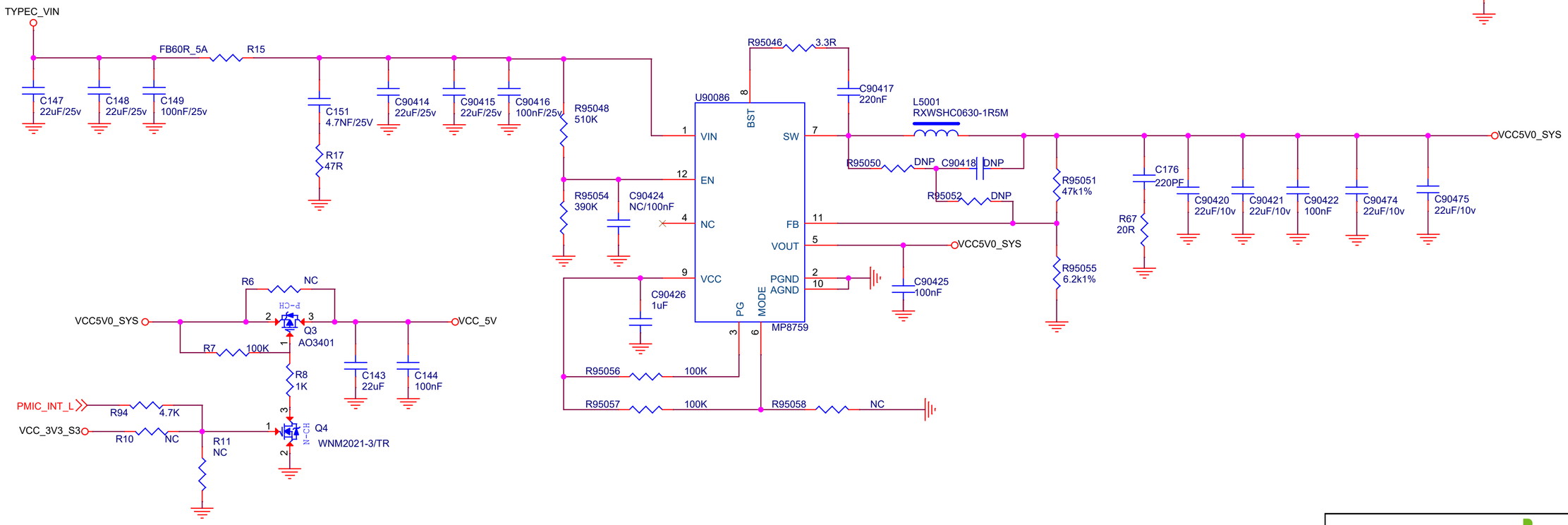




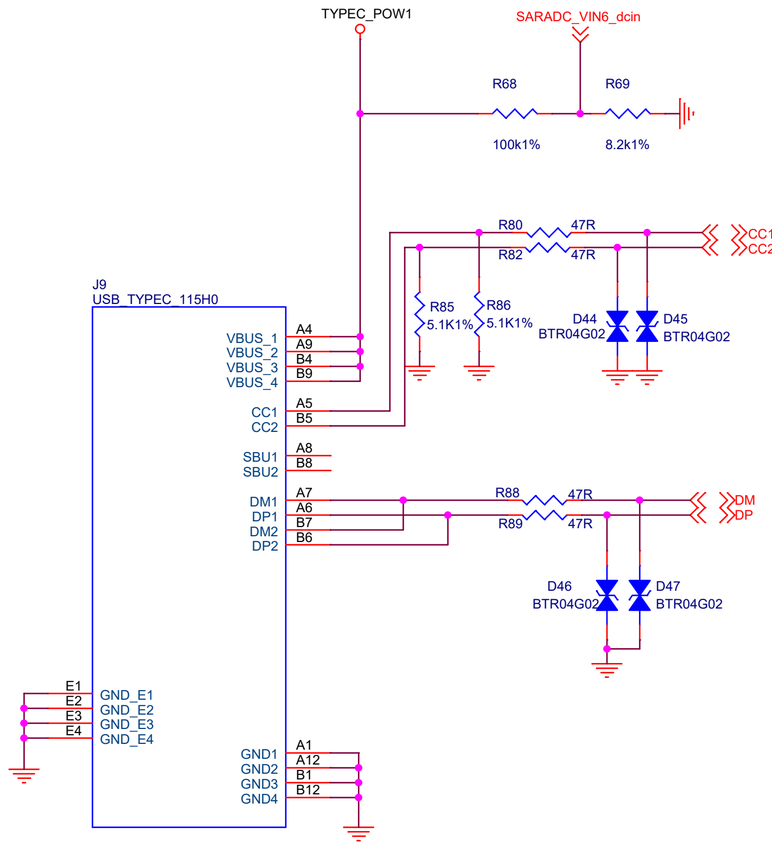
Size A2	Title: ROCK 5B+		REV 1.2
	Page Name: 31.Power-PMIC_RK806-1		
Date: Thursday, August 15, 2024	Sheet 31 of 33		

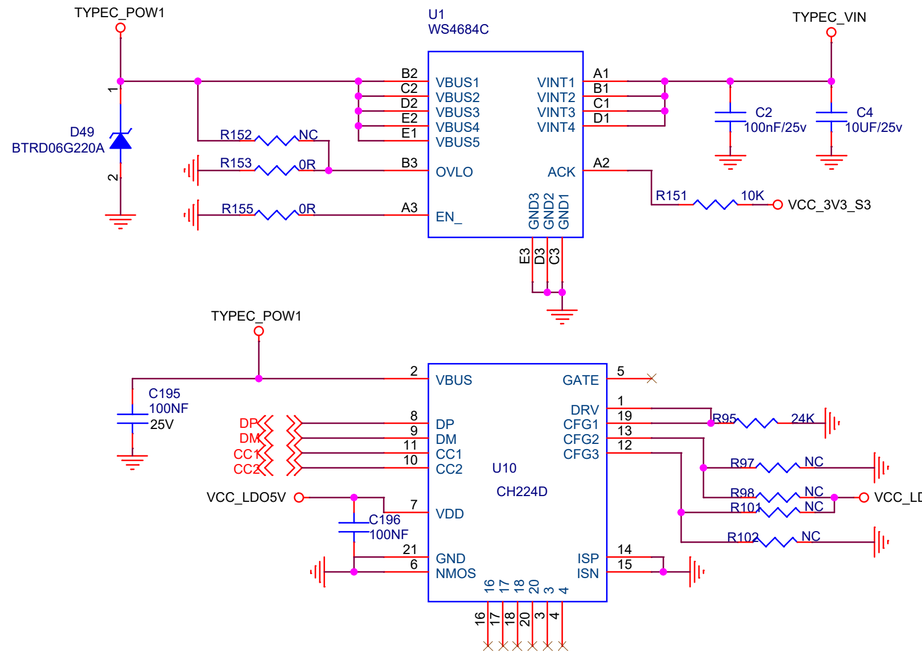


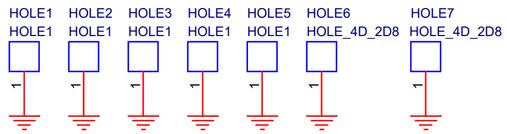




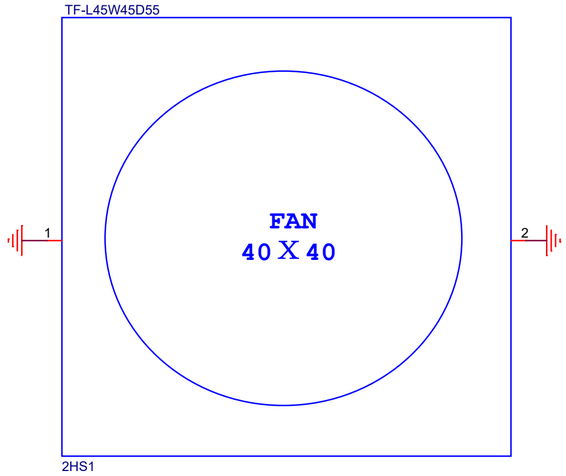
Size A3	Title: ROCK 5B+		REV 1.2
	Page Name: 32.Power_DC 5V		
Date: Thursday, August 15, 2024		Sheet 32 of 33	











Size	Title:	ROCK 5B+	REV
Custom	Page Name:	33.Power TYPEC IN	1.2
Date:	Thursday, August 15, 2024	Sheet 33 of 33	